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Empirical Analysis of Fragility, Robustness, and Antifragility in Investment Portfolios

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Introduction

In recent centuries, with the advent of industrialization, globalization, information technology and, more recently, artificial intelligence, the reality that surrounds us has become increasingly complex. The increase in complexity imposes equally complex decisions, the consequences of which (notwithstanding the technological progress of analytical tools) prove increasingly difficult to predict and often of increasing magnitude. At times, the very complexity of reality "presents us with the bill," generating unforeseen and potentially catastrophic outcomes. The inability to estimate with precision the magnitude and the probability of damages deriving from such events imposes a reflection on the definition of the concept of risk itself. Within this perspective is situated the work carried out by the scholar Nassim Nicholas Taleb. Taleb, a mathematician expert in probability theory and financial mathematics, has written several mathematical articles, and a series of popular essays, in which he provides an innovative perspective on the approach to risk management and uncertainty in complex systems. According to him, the fundamental change consists in shifting the attention from the attempt to predict the probability of an extreme event to the analysis of its consequences, for the purpose of preparing strategies capable of limiting or even exploiting its impact. Such a vision, rather than on the identification of the cause of the phenomenon, concentrates on the effects that it produces. From this framework derive the concepts of *fragility*, *robustness*, and *antifragility*, which Taleb formalizes also from a mathematical point of view. The advantage of such an approach resides in not being based on simplifying hypotheses or on predefined probability distributions, but on the empirical observation of real data. The disadvantage is the necessity of having at one's disposal a large sample of data to fully understand the properties of the phenomenon. To overcome such a limit, Taleb proposes a heuristic capable of identifying in a qualitative way fragile systems and antifragile ones. In this thesis, the principal ideas presented by Taleb in his writings are analysed, together with the mathematical definitions of the concepts of fragility, robustness, and antifragility, with the objective of verifying their applicability to the financial context. In particular, the aim is to evaluate the effectiveness of such measures as alternatives to traditional definitions of risk.

Chapter 1 introduces the theoretical framework of reference, retracing the historical development of probabilistic thought, from Gaussian models to the contributions of Taleb. The concepts of *Mediocristan* and *Extremistan*, *power laws*, the phenomenon of *Black Swans*, and the "fourth quadrant" are discussed, with a final focus on financial returns and portfolio theory.

Chapter 2 delves into the heuristic proposed by Taleb to identify fragility and antifragility, together with the relative mathematical formulations. Furthermore, the principal traditional measures of risk and performance of portfolios are examined, finally discussing the role of options as antifragile instruments and the strategies that derive from them.

Chapter 3 presents the empirical application of the theoretical definitions to different investment strategies. The case of passive investment in an equity index and the effect of periodic hedging with Put options for protection from extreme events is analysed. The analysis concludes with the study of various *Barbell* portfolios, with the objective of evaluating the fragility of markets, the antifragile behaviour of options, and the robustness of the proposed Barbell strategies.

1. Uncertainty and risk in financial markets: from classical models to black swans

For several centuries, philosophers, mathematicians, and physicists have studied and questioned the concept of probability and the forecasting of events. The term statistics derives from the German *Statistik*, which literally means the science of the state. John Graunt, an English intellectual, published the first demographic study based on real numbers in 1662. The initial objective was therefore to collect data and conduct analyses for governmental purposes. In parallel, a second important element for statistics emerged through the first studies on the calculation of probabilities in gambling. In this sense, the exchange of letters in 1654 between Blaise Pascal and Pierre de Fermat is considered the birth certificate of the mathematical theory of probability. Subsequently, in the early 1700s, Jacob Bernoulli published the treatise “*Ars Conjectandi*” in which he introduced the law of large numbers. In 1763, “*An Essay towards solving a Problem in the Doctrine of Chances*” was published by Thomas Bayes. In this treatise we find Bayes' theorem, important because it introduces into the calculation of probabilities a mechanism of updating conditioned on new information. According to Bayes, probability is the degree of confidence or belief that an individual has regarding a certain event; this probability is therefore variable and subjective. The concept is still relevant today because it clashes with the simplifying hypothesis, often used in economics and finance, according to which rational individuals converge toward the same probabilities. In this period, the studies of Abraham de Moivre, Gauss, and Laplace also developed, which would lead to the development of the normal curve and the central limit theorem. In this context, there emerged a deterministic version of the calculation of probabilities that characterized studies on probability until recent times. We find a representation of this current of thought in Laplace's treatise “*Essai philosophique sur les probabilités*” published in 1814, according to which if we were able to know the position and the velocity of every particle in the universe, we would be able to predict the future with certainty. To date, this belief is considered superseded: in the early 1900s, the first models were born that include randomness as an intrinsic property of the model. One of the first formalizations derived from an application in the field of finance, particularly in the modelling of prices in markets. In his doctoral thesis, the mathematician Bachelier determined the probability of changes in prices through the Chapman–Kolmogorov equation, recognizing that what we now call a Wiener process or Brownian motion satisfies the diffusion equation. It is here that, for the first time, economics meets physics. The diffusion equation in physics describes the macroscopic behaviour resulting from the random

movements of microparticles caused by collisions. This concept was later rediscovered by Einstein in 1905 and reported in his paper on Brownian motion “Investigations on the theory of the Brownian movement.” The assumption of normal returns was later modified, assuming that prices follow a log-normal distribution, while returns follow a normal distribution. However, empirical evidence demonstrates that the tails of the distribution of returns are wider compared to those predicted by Brownian motion and that the standard deviation changes over time. It is in this context that physicists have inserted themselves. The contribution of physics to economics consists in utilizing models that were not exactly designed to model markets and are not based on the Gaussian. In this way, they add a different point of view to the study of the subject. In recent years, physics has made progress in the study of statistical mechanics, non-linear dynamics, and disordered systems; in these fields, time series, scalability, and power laws are present. The current interpretation of the underlying physics of these phenomena is often obtained using these concepts. Therefore, the contribution of physicists is substantiated in the application of these methods in the study of markets. This is proving particularly useful in modelling extreme risks for which the Gaussian and its variants are not a good approximation of reality. The innovation brought is therefore particularly useful in terms of risk management. In the remainder of this chapter, we will analyse the normal distribution and its limits, discuss the power law and its utility, and finally conclude with a focus on markets and portfolio theory.

The Normal Distribution: origin and characteristics

In this paragraph we seek to explain what normality is in statistics, starting therefore from the etymology to understand what it means, how it is defined, and through what means it is identified. The term “normal” has Latin origin and specifically derives from “norma,” which means square, referring to the square used to trace right angles. In a figurative sense, it means instead “rule”; therefore, the adjective “normal” defines that which is compliant with a determined principle. The criterion utilized varies according to the context in which one wishes to study a certain phenomenon. For example, when we say according to the law, we wish to indicate that an act is congruent with what is provided for by the law. When we speak of a normal behaviour, we wish in reality to say that this is commonly accepted, that is, customary. In statistics, normality is described by a probability density function, which has a precise graphical representation. Figure 1 graphically describes the bell curve, so named because of its shape. Observations that fall within the central part of the curve are considered normal, while those at the extremes are deemed abnormal, insofar as they are exceptionally rare, with a low probability of occurrence associated with them. Applying the concept of statistical normality outside of its own perimeter might therefore not be correct, as it can assume

different meanings and connotations. The first to approach this theme was Abraham de Moivre who, in 1733, in an article on gambling¹, demonstrated how the binomial distribution, as the number of trials increases, tends toward a normal distribution. The concept was then taken up by Carl Friedrich Gauss, who made it famous in the work “*Theoria motus corporum coelestium*”² published in 1809. In this treatise on mathematical astronomy, Gauss highlights the normal curve, which indeed will take his name, becoming known also as the Gaussian. The Gaussian probability density function for a random variable with mean μ and variance σ^2 is the following:

Eq. 1
$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Gauss utilized this discovery as a tool to model measurement errors, applying it to his studies of astronomy. It was then Laplace who elaborated a broader theory on the calculation of probabilities in his work “*Théorie analytique des probabilités* (2nd ed.)”³ of 1812. In this work, we find one of the first formulations of the central limit theorem. Laplace shows that, as the number of trials increases, the distribution of the outcomes of a sequence of independent and identically distributed (i.i.d.) trials converges to the normal distribution. Furthermore, he introduced a systematic approach to the calculation of probabilities based on integral calculus, extending and formalizing what had been previously done by Bernoulli and de Moivre. This treatise is historically important because it provided the fundamental tools and theories upon which statistics and econometrics are still based today.

Before delving into the reasoning at the base of this tool, let us analyse the principal characteristics of this curve. Looking at Figure 1, the first thing we notice is the symmetry with respect to the centre. The central value is represented by the mean. The red lines have been drawn in correspondence with the standard deviations. These tell us how much an observed value diverges from the mean and, consequently, allow us to establish the “rarity” of that observation. The rule is as follows: values within 1σ have a probability of being observed equal to 68%, values within 2σ equal to approximately 95%, while within 3σ equal to 99.7%.

Consequently, values beyond three standard deviations are very rare and should with difficulty be observed. Unfortunately, the Gaussian has demonstrated that it significantly underestimates the

¹ Abraham de Moivre, *Approximatio ad Summam Terminorum Binomii in Seriem Expansi* (1733), in which he showed for the first time that the binomial distribution, as the number of trials increases, can be approximated by a continuous bell-shaped curve, anticipating the concept of the normal distribution.

² Carl F. Gauss, *Theoria motus corporum coelestium in sectionibus conicis solem ambientium* (1809), Gauss studies measurement errors in the movement of planets introducing the method of least squares and applied the Gaussian function as the law of errors of observation.

³ Pierre-Simon Laplace, *Théorie analytique des probabilités*, 2^a ed., Paris, Courcier, (1812).

probabilities of occurrence of these events. Some phenomena assume distributions characterized by thicker tails; this is one of the principal biases of the Gaussian. We will see later that the normal distribution is not always a good approximation of reality. We will seek to identify the cases in which it is appropriate to use it from those in which it is not. Two other parameters that do not emerge intuitively by looking at the graph, but which are important for describing the Gaussian are: skewness and kurtosis. Skewness indicates to us how much a distribution is symmetrical with respect to the mean, that is, if it is shifted toward the left or toward the right; in the normal, it assumes a value of zero. Kurtosis, instead, determines the shape of the curves. A high kurtosis indicates curves with a very pronounced centre and thicker tails, while a low kurtosis indicates flatter distributions with thin tails. In the normal distribution, the kurtosis has a value equal to three.

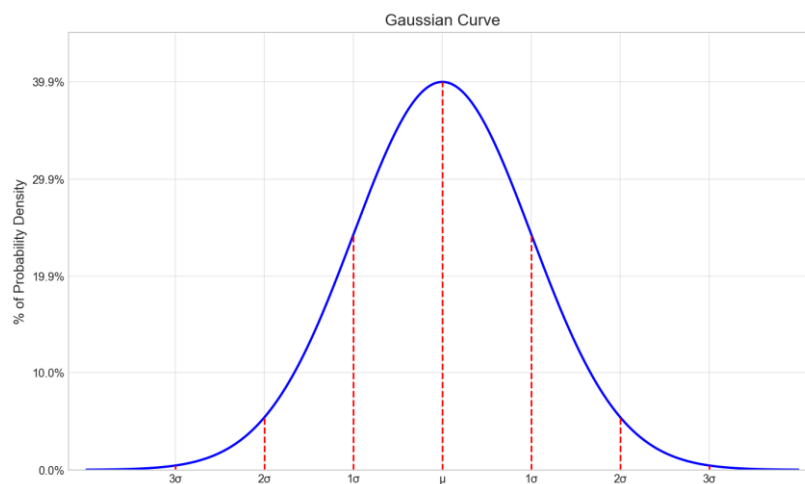


Figure 1. The Gaussian curve

But why does the Gaussian assume precisely this form? To explain in a simple and intuitive manner the logic with which the bell curve is formed, Francis Galton⁴ invented the Galton box. This is an object divided into two sections: an upper one containing small balls, and a lower one containing cylinder arranged in the shape of an inverted tree. The functioning of the game is illustrated by Figure 2.

⁴ Francis Galton (1822-1911) was a versatile British scientist and intellectual, known for his pioneering contributions in various fields. A cousin of Charles Darwin, he distinguished himself as an explorer, meteorologist, and pioneer of statistics, where he introduced fundamental concepts such as correlation and regression.

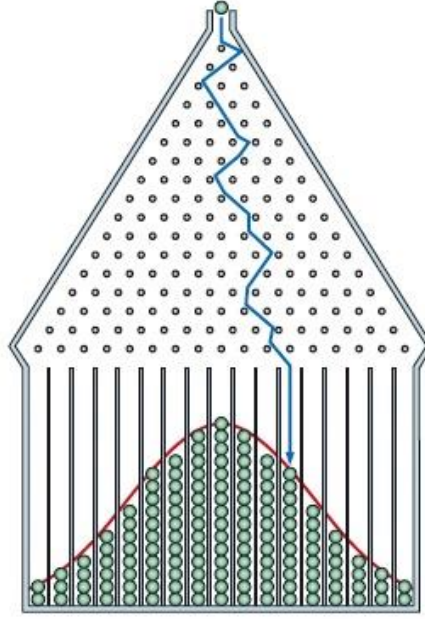


Figure 2. The Galton board

The ball represents the random variable. Each time we let a ball fall into the box, it will pass all the cylinders and deposit itself in one of the rectangles at the bottom. Each cylinder represents, metaphorically, a choice for the ball with only two possible alternatives: going to the right or to the left. The probabilities are the same for both directions. The game demonstrates that by throwing a high number of balls, these deposit themselves assuming the shape of the bell curve. Why? Let us observe what happens at the extremes. For a ball to go into the last rectangle, it would have to fall continuously to the right. The combination of events that makes such a result possible is only one. In the figure there are 17 cylinders; therefore, the ball must go to the right 17 times. Conversely, to go into the first position, the ball must go to the left 17 times. What can we say, instead, of the events that are found in the centre? To answer, we must reformulate the question in this way: what choices can the ball make to end up in the central compartment? In this case, there are more possible paths. For example, it could go once to the right and once to the left for 17 times. Or twice to the right and twice to the left until arriving at the bottom. Combinatorial calculus allows us to determine in a simpler way the total number of possible paths through the binomial coefficient:

$$\text{Eq. 2} \quad C(n, k) = \binom{n}{k} = \frac{n!}{k! (n - k)!}$$

The binomial coefficient calculates the number of ways in which distinct elements can be chosen from distinct elements, regardless of the order. In our case, is equal to 17, while is the number of

times that the ball deviates toward the right and determines, therefore, the position in which it will fall. Let us calculate, with the binomial coefficient, the combinations for.

$$C(17,0) = \frac{17!}{0!(17-0)!} = 1 \quad C(17,9) = \frac{17!}{9!(17-9)!} = 24\,310 \quad C(17,17) = \frac{17!}{17!(17-17)!} = 1$$

That which we had previously stated only logically now finds a quantitative confirmation. We notice, therefore, that there exists only one combination that leads the ball into the first and into the last column, while 24,310 combinations that lead it into the middle. We can now calculate the probabilities associated with such combinations through the binomial distribution:

$$p(k; n, p) = \binom{n}{k} \cdot p^k \cdot (1 - p)^{n-k}$$

p represents the probability of success, that is, the probability that the event occurs times out of. Let us calculate such probability for the binomial coefficients calculated previously:

$$\begin{aligned} p(0; 17, 0.5) &= 1 \cdot 1 \cdot 0.5^{17} = 0.00076\% \\ p(9; 17, 0.5) &= 24310 \cdot 0.5^9 \cdot 0.5^8 = 18.54\% \\ p(17; 17, 0.5) &= 1 \cdot 1 \cdot 0.5^{17} = 0.00076\% \end{aligned}$$

At this point it is clear, also from a mathematical point of view, why the normal distribution assumes this form and why the events closer to the centre are more probable, so much so that they can be considered “normal.” On the contrary, events on the tails of the distribution are often treated as exceptional, precisely because of their low probability of occurrence. To calculate these probabilities, we used the binomial distribution. But as the Galton box already clearly shows us, the binomial distribution, as (levels) increases, approaches the normal distribution. Laplace formalizes this mechanism of approximation to the normal for the binomial distribution through the central limit theorem. The theorem states that the distribution of the sum of a large number of independent and identically distributed random variables, centered and normalized, tends to be distributed as a standard normal with mean 0 and variance 1. Subsequently, other formulations of the theorem, such as that of Lindeberg and Paul Lévy⁵, have extended the result, allowing for the relaxation of the hypothesis of identical distribution: the variables do not necessarily have to come from the same law,

⁵ Paul Lévy, *Calcul des probabilités* (1925), his work in which he generalizes the Central Limit Theorem to non-identically distributed random variables, introducing concepts such as the “Lindeberg condition.”

provided that they satisfy weaker regularity conditions (for example, the Lindeberg condition). At the base, however, there is an important assumption, namely that the observations must be independent. This implies that if the observations are in some way correlated with each other, and thus not independent, the approximation to the normal of such a distribution would lead to distorted and inconsistent results. In this paragraph, we have examined the concept of normality in statistics, its origins, and its logic. In the next paragraph, we will take a look at modern reality: we will examine the importance of the Gaussian, we will recognize its limits, and we will seek to define a perimeter of applicability using metaphorical models and the ideas proposed by Nassim Nicholas Taleb.

Mediocristan and Extremistan: Normal and Power Law in Comparison

The Gaussian curve, the law of large numbers, the central limit theorem, and linear regression represent the foundations upon which statistics and econometrics are based. In this paragraph, utilizing Taleb's metaphors, we will seek to understand when these tools can be utilized, deferring the discussion on financial application to the next paragraph. The objective is therefore to identify the situations in which the Gaussian is capable of correctly representing reality. To do so, we introduce the metaphorical concepts of *Mediocristan* and *Extremistan*, of which it is possible to read a detailed description within Taleb's books.

What is Mediocristan? Mediocristan is an imaginary world governed by simple laws, which make everything that inhabits it predictable and easy to describe. Fifty or so observations would be sufficient to adequately describe a phenomenon of Mediocristan. In this environment, surprises count for little, because events contribute only from a collective point of view. Quoting Taleb:

“The supreme law of Mediocristan could be expressed thus: when the sample is large, no single case can significantly change the aggregate, that is, the total. The largest datum may be impressive, but it remains insignificant relative to the sum.”⁶

Examples of some phenomena belonging to Mediocristan are the height of a man or a woman, body weight, intelligence quotient (IQ), the number of victims of road accidents in a year, the birth rate, minutes spent on the telephone in a day, the number of books read in a year by a person. Single extreme observations of these phenomena do not contribute to modifying the mean in a relevant manner. For example, in the case of intelligence quotient, if out of a sample of one hundred people, one were to result with an IQ equal to one hundred forty, such an observation would not modify the

⁶ Taleb N.N., *Il Cigno Nero* (2007), Il Saggiatore, Milano, p. 54.

mean of the sample. In this world, the central limit theorem and the bell curve correctly represent reality. This also explains to us why the Gaussian is important, that is, because it emerges spontaneously from the observation of many natural phenomena.

Let us now move to the other side of the coin. What is Extremistan? Extremistan is the exact opposite: it is a complicated world in which events do not repeat themselves in an exact manner over time. There are no laws that govern it; in a certain sense, it is as if it were governed by chaos. The elements that compose it can be connected to others, bound by non-linear relationships that are difficult to identify. Such characteristics make these phenomena difficult to predict, and extreme events can have a great impact on the population. Moreover, unlike the events of Mediocristan, numerous observations are necessary to make the real properties of a phenomenon emerge. I report some examples of cases that belong to Extremistan: financial markets, the distribution of wealth, the success of artists and writers (number of CDs or books sold), technological innovations, wars, pandemics, rare natural catastrophes. From this list, we can appreciate the importance of such events, upon whose occurrence correspond great changes for the lives of people, communities, and nations. To better understand why Extremistan is important, it is necessary to introduce the definition of the Black Swan. A Black Swan is an unpredictable, unexpected event, explainable only in retrospect, which has an enormous impact on reality. By definition, these phenomena cannot originate in Mediocristan, but on the contrary, Extremistan is the environment suited for their development. Although there does not exist a probability distribution capable of fully capturing these phenomena, the power law can make the probability of extreme events understandable, unlike the Gaussian which would render them almost impossible. This approach effectively transforms some “Black Swans” into “Grey Swans,” since a mathematical model allows them to be rendered partially analysable. I report a numerical example⁷ taken from “The Black Swan” which clearly explains the concept:

Scalable wealth distribution (power law)

People with a net worth exceeding 1 million euros: 1 in 62.5

Exceeding 2 million euros: 1 in 250

Exceeding 4 million euros: 1 in 1,000

Exceeding 8 million euros: 1 in 4,000

Exceeding 16 million euros: 1 in 16,000

Exceeding 32 million euros: 1 in 64,000

Exceeding 320 million euros: 1 in 6,400,000

⁷ Taleb N.N., *Il Cigno Nero* (2007), Il Saggiatore, Milano, pp. 245, 246.

We notice that as wealth increases, the probabilities of encountering a wealthy person decrease at a constant rate (1.6% - 0.4% - 0.1% - 0.025% - 0.00625% - 0.00156% - 0.00001562%). Let us see what these probabilities would be if they were estimated with the Gaussian.

Distribution of wealth assuming the Gaussian law

People with a net worth exceeding 1 million euros: 1 in 63

Exceeding 2 million euros: 1 in 127,000

Exceeding 3 million euros: 1 in 14,000,000,000

Exceeding 4 million euros: 1 in 860,000,000,000,000,000

Exceeding 8 million euros: 1 in 16,000,000,000,000,000,000,000,000,000,000,000,000,000,000

Exceeding 16 million euros: 1 in incalculable

The probabilities of encountering a wealthy person are practically null already starting from 4 million euros. The problem is that in reality, there are relatively many more of these people. We notice that the probabilities decrease in an exponential manner, and this is the principal limit of the Gaussian distribution applied to a phenomenon belonging to Extremistan. The power law makes the possibility of encountering extreme observations seem real, however improbable. This does not mean that the exact probability of encountering a man with a net worth exceeding 320 million is 1 in 6,400,000. It allows us at least to consider that this probability exists. In practice, the power law permits the partial resolution of the problem of the “fat tails” of the normal distribution and its variants.



Figure 3. Wealth distribution assuming the normal law

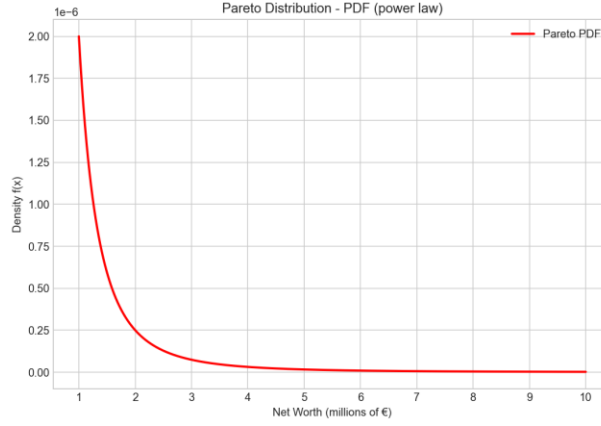


Figure 4. Wealth distribution assuming the power law

Let us now see how the probabilities of our example were calculated through the use of the power law and how, starting from a few data points, we can construct a probability distribution. Let us imagine having a sample of n people with n different incomes. The power law distribution (also called the Pareto Distribution) can be expressed in its simplest form as:

$$\text{Eq. 3} \quad p(x) \propto L(x)x^{-\alpha}$$

This probability density function tells us that the probability of observing large values of decreases as a power law. The symbol $\propto$ is a multiplicative factor that serves to normalize the pdf. α is the exponent. $L(x)$ is a slowly varying function, such that:

$$\text{Eq. 4} \quad \lim_{x \rightarrow \infty} \frac{L(tx)}{L(x)} = 1$$

When $L(x)$ is constant, the distribution becomes:

$$\text{Eq. 5} \quad p(x) = C x^{-\alpha}$$

Now the pdf has a precise form. When $L(x)$ is constant, the shape of the distribution does not change and is subject only to changes of scale (our x). This means that $L(x) = C$ is the condition that renders the pdf scalable and therefore self-similar⁸. If we calculate the integral of $p(x)$ between 0 and $+\infty$,

⁸ Self-similar: an object, a distribution, or a function that maintains the same shape when it is enlarged or reduced in scale.

we notice that the distribution diverges when tends toward zero for positive values of α . We must impose a minimum value x_{min} . Therefore, when x is a continuous variable:

$$\text{Eq. 6} \quad C = 1 / \int_{x_{min}}^{\infty} x^{-\alpha} dx$$

from which:

$$\text{Eq. 7} \quad p(x) = \frac{\alpha - 1}{x_{min}} \left(\frac{x}{x_{min}} \right)^{-\alpha}$$

In our example α is equal to 2, $x_{min} = 1\,000\,000$. Assuming that the probability (incidence) of observing a millionaire, man is 1 in 62.5 (1.6%). Let us calculate the probabilities for the other thresholds by scaling the $p(x)$ by the empirical probability 0.016:

Table 1: Power law probability

	$p(x)$	$P(x)$	Incidence
For 1 million	$1^{-2} = 1$	$0.016 \times 1 = 0.016$	1 in 62.5
For 2 million	$2^{-2} = 0.25$	$0.016 \times 0.25 = 0.004$	1 in 250
For 4 million	$4^{-2} = 0.0625$	$0.016 \times 0.0625 = 0.001$	1 in 1 000
For 8 million	$8^{-2} = 0.0156$	$0.016 \times 0.0156 = 0.00025$	1 in 4 000
For 16 million	$16^{-2} = 0.0039$	$0.016 \times 0.0039 = 0.000065$	1 in 16 000

The parameter α , which in our case was 2, is generally the result of an estimation. Estimates of α can be based on historical data or on theoretical models. I report now a table of estimates of α for various phenomena that follow the power law, extracted from *The Black Swan*⁹:

Table 2: Power law exponents for various phenomena

Phenomenon	Hypothesis of exponent (vague approximation)
Frequency of use of words	1.2
Number of visits to a website	1.4
Number of books sold in the United States	1.5
Telephone calls received	1.22

⁹ Taleb N.N., *Il Cigno Nero* (2007), Il Saggiatore, Milano, p. 275, sotto fonte M.E.J Newman, 2005

Magnitude of earthquakes	2.8
Diameter of moon craters	2.14
Intensity of solar flare	0.8
Intensity of wars	0.8
Net income of Americans	1.1
Number of people per surname	1
Population of American cities	1.3
Market movements	3 (or lower)
Size of companies	1.5
People killed in terrorist attacks	2 (or lower)

The estimation of the exponent is typically overestimated; it induces us to think that rare events are less probable than they really are. It is necessary to keep in mind that even small estimation errors produce important effects on the probabilities $p(x)$, as illustrated by the following table.

Table 3: Variations in power law probabilities for changes in the estimation of the exponent

Alpha	1M	2M	4M	8M	16M	32M	320M
1.0	1.0	0.500000	0.250000	0.125000	0.062500	0.031250	0.003125
1.1	1.0	0.466516	0.217638	0.101532	0.047366	0.022097	0.001755
1.3	1.0	0.406126	0.164938	0.066986	0.027205	0.011049	0.000554
1.5	1.0	0.353553	0.125000	0.044194	0.015625	0.005524	0.000175
1.7	1.0	0.307786	0.094732	0.029157	0.008974	0.002762	0.000055
2.0	1.0	0.250000	0.062500	0.015625	0.003906	0.000977	0.000010
2.5	1.0	0.176777	0.031250	0.005524	0.000977	0.000173	0.000001
3.0	1.0	0.125000	0.015625	0.001953	0.000244	0.000031	0.000000
4.0	1.0	0.062500	0.003906	0.000244	0.000015	0.000001	0.000000
5.0	1.0	0.031250	0.000977	0.000031	0.000001	0.000000	0.000000
6.0	1.0	0.015625	0.000244	0.000004	0.000000	0.000000	0.000000

What we notice is that the probabilities of extreme events substantially converge to those expressed by the normal for high values of α . A small variation from 1.0 to 1.1 reduces by almost half the probability of observing a net worth exceeding 320 million euros. Another particular property of the Pareto distribution concerns the second moment. The moment of order m is given by the following formula:

Eq. 8
$$\langle x^m \rangle = \int_{x_{\min}}^{\infty} x^m p(x) dx = \frac{\alpha - 1}{\alpha - 1 - m} x_{\min}^m$$

The expected value for the pdf of the power law is defined only when $\alpha < 2$, while the variance is finite only if $\alpha > 3$. Therefore, when $2 < \alpha < 3$, the mean is finite, but the variance is infinite.

To conclude the discussion, I report another table that briefly illustrates the distinctive characteristics of Mediocristan and Extremistan; for a more in-depth discussion, I refer to the reading of *The Black Swan*. In the next paragraph, we will examine the case of financial markets, seeking to understand what, according to Taleb, led to the 2008 crisis, and the concept of the fourth quadrant.

Table 4: Differences between Mediocristan and Extremistan

Mediocristan	Extremistan
Non-scalable	Scalable
Type 1 or mild randomness	Type 2 or wild (or even super-wild) randomness
The most typical member is mediocre	The most typical is a giant or a dwarf; that is, a typical member does not exist
Winners receive a small slice of the pie	Winners receive almost all of the pie
Example: audience of an opera singer before the invention of the gramophone	Audience of a contemporary artist
More common in our ancestral environment	More common in our modern environment
Inaccessible to the Black Swan	Vulnerable to the Black Swan
Subject to gravity	The datum is not subject to physical limits
Corresponds (generally) to physical quantities, for example stature	Corresponds to numbers, for example wealth
As close to utopian equality as reality can be	Dominated by an extreme inequality of the winner-take-all type
The total is not determined by a single case or by a single observation	The total is dominated by a few extreme events
If one observes for a while, one understands how things stand	A long time is needed to understand how things stand
Tyranny of the collective	Tyranny of the accidental

Easy to predict based on what is seen and to extend to what is not seen	Difficult to predict based on information from the past
History creeps	History jumps
Events are distributed according to the bell curve or one of its variants	The distribution consists of Mandelbrotian Grey Swans (scientifically manageable) or totally unmanageable Black Swans ¹⁰

Fat tails, black swans, and the fourth quadrant in financial markets

In this paragraph, we will take a look at financial markets, seeking to understand what the criticalities of the Gaussian are in financial markets. We will address the criticalities of the bell curve and the problems linked to the estimation of risk through standard deviation. We will conclude the discussion by introducing the concept of the fourth quadrant presented by Taleb as a useful tool to understand when it is possible to rely on mathematical models and when, instead, these should not be taken into consideration.

The principal issue, that from which subsequent considerations on risks also derive, concerns the distribution of the returns of financial instruments. Normally, it is assumed that returns, or rather log-returns, follow a normal distribution. This assumption, however, seems not to be verified empirically. To delve into the matter, the weekly returns of the S&P 500 from 1950 to the present have been examined. The table illustrates the changes in the distributions of returns as time increases:

Table 2: Variations of the moments of the distribution of returns over time

Period	Mean	Std. Deviation	Skewness	Kurtosis	Extreme Obs.
1950 - 1987	0.001374	0.018557	-0.155125	2.946264	25
1950 - 1997	0.001537	0.018648	-0.362317	3.606188	28
1950 - 2008	0.001486	0.019614	-0.408291	3.548872	34
1950 - 2020	0.001435	0.020580	-0.599054	5.725715	49
1950 - 2025	0.001503	0.021213	-0.645054	6.184852	53

¹⁰ Taleb N.N., *Il Cigno Nero* (2007), Il Saggiatore, Milano, p. 57

Each period was chosen so that it preceded a crisis; in this way, it is possible to analyse the changes that occurred in the distributions from one crisis to another. From the table, it is possible to notice that the mean of the returns remains stably around zero; negative and positive movements balance each other out in magnitude. The extreme observations of negative returns are more numerous than the positive ones. At the same time, the total number of positive returns is greater than the negative ones, suggesting that the downturns are more violent and more often exceed 3 standard deviations. The magnitude of the negative and positive movements balances out, giving rise to a distribution with a mean almost equal to zero. From the subsequent graph, we can furthermore notice how these variations tend to concentrate. We notice, moreover, that following extraordinary negative returns, extraordinary positive returns are observed.



Figure 5. Extreme returns in the S&P 500

Regarding the standard deviation, we can notice that it has consistently increased. The crises have added new variability to the distribution. In more recent periods, the increase has been greater, perhaps due to the increase in volumes and the transition to digital markets. The distribution of returns, over time, has increasingly shifted toward the left. The left tail has lengthened. The kurtosis has almost always increased, with the sole exception of the dot-com crisis. The 2008 crisis had the greatest impact on kurtosis. Furthermore, we notice that extreme observations have never decreased. This means that what was an extreme return in 1987 was also extreme in 2025, notwithstanding the increase in kurtosis. I report the graph that takes into consideration the distribution of returns for the entire period considered, 1950–2025.

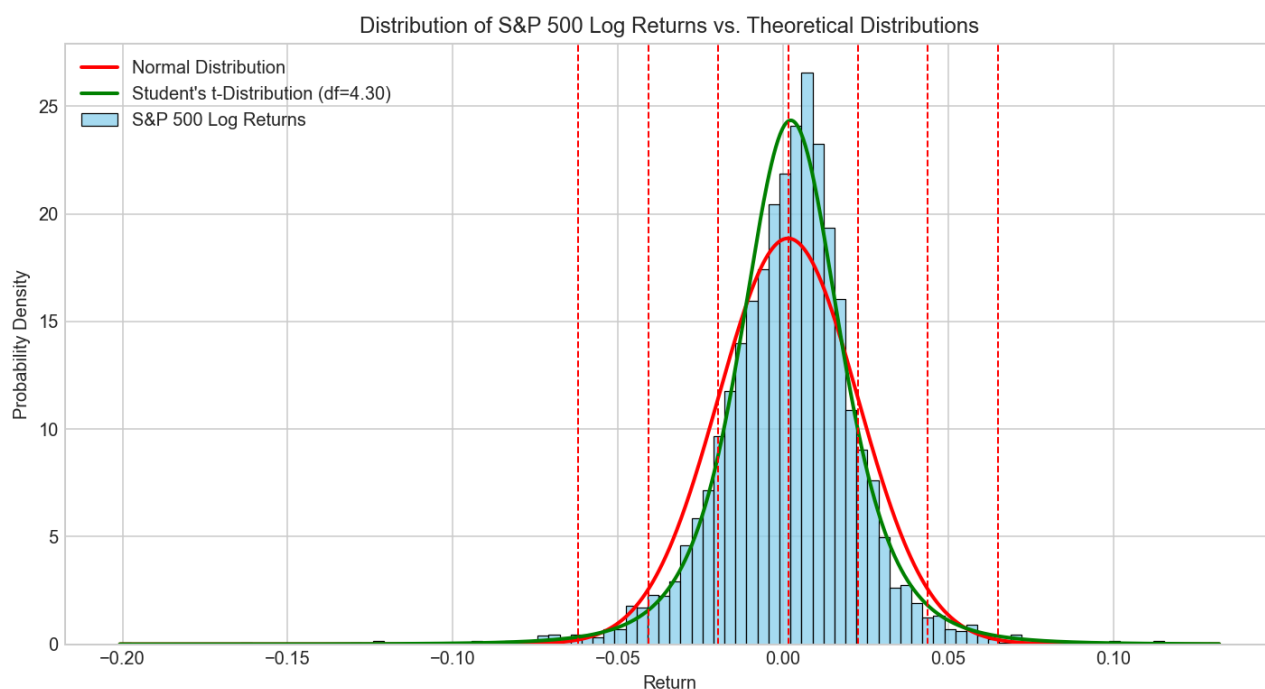


Figure 6. Distribution of S&P 500 returns

In light blue we have the empirical distribution of log-returns, in red the normal distribution, and in green the student's t-distribution with 4.29 degrees of freedom. From the image we can notice that the Gaussian does not adapt well to the observed returns. The dashed red lines represent the standard deviations. Negative returns lower than -0.062 exceed three standard deviations and position themselves on the left tail; in 75 years, 35 of these events have been observed. The worst weekly close examined was observed on 10/08/2008 with a negative return of approximately -20% . According to the Gaussian, this event has a probability of occurring equal to $7.690045e-22$. The mean of the probabilities associated with negative extreme events is equal to 0.036% ; in practice, we should observe events of this type approximately once every 53 years. Meanwhile, returns that exceed 6.5% position themselves on the right tail; of these, we have observed 18. In total, there are 53 extreme observations not explainable through the Gaussian. On the contrary, the Student's t-distribution would seem to correctly represent the central part, but it nonetheless underestimates the probabilities of events that go beyond 3 standard deviations. The hypothesis is that extreme returns are better represented by the power law.

The preceding analysis demonstrates empirically that returns do not follow the Gaussian. This implies that the standard deviation is not an adequate measure to measure the risk of a financial asset, or, to put it better, it is not suited to measure risk in conditions of extreme uncertainty. For this reason, in "normal" periods characterized by low volatility, the standard deviation can seem an adequate representation. The problem is that it is not possible to predict a crisis. Therefore, utilizing the

standard deviation as a measure means implicitly assuming that there is no risk of crisis (or that it is extremely improbable). Taleb maintains that people confuse periods of low volatility with quiet periods and not with journeys toward Extremistan. The fact is that the market could pass through periods of tranquillity, but the importance of these moments is relatively low if compared to moments of crisis. The impact of a rare event is much greater in reality. The point is: if the Gaussian does not allow us to imagine these events, does the power law allow us to do so? The answer is partial. The power law would seem to better represent extreme returns, rendering them at least possible. But even this distribution is subject to limits and is not suited to represent extremely rare events. In practice, a correct measure of risk does not exist.

“It is not possible to measure risk in the same way in which we can measure temperature with a thermometer.”

Mathematical models do not allow us to make exact predictions nor to represent all of reality. What we find in models is a simplification of that which surrounds us, based on our current knowledge. There are sources of uncertainty that we do not know and perhaps could never know. One cannot escape from “black swans.” The power law allows us to transform some of these into “grey swans.” Taleb describes three problems for which mathematical models are not capable of correctly representing reality and its risks.

The first problem, self-reference, consists in the fact that to estimate the probability distribution I need past data. But to estimate future behaviour starting from historical data, a probability distribution is necessary that is capable of verifying the adequacy of these data. In practice, a rule is needed that establishes a priori which are the suitable tools for analysing a phenomenon. But in the presence of rare events, I will be able to collect only a few data points to elaborate a model. Therefore, the prediction error of the model will be greater as the rarity of a phenomenon increases. When we find ourselves in Mediocristan, the consequences of a rare event are of little importance (because the collective effect prevails). In Extremistan, where avalanche effects occur, the consequences multiply. Even a small error in the model can generate an immense effect.

The second problem is linked to the asymptotic properties of the models. In the asymptote, mathematical models result as exact/rigorous, but in reality, we live in the pre-asymptote. Models hypothesize that in the middle nothing new happens that can modify our choices, but this is not real. We are almost never able to reach the asymptote.

The third problem concerns inversion. That is, the attempt to trace back from the knowledge of effects to causes. It is like trying to understand from a puddle the shape of the ice solid that generated it. There does not exist a finite number of answers. It is easier to do the opposite—that is, from the ice, to estimate the size of the puddle. We know the origins of crises only after they occur. We are unable to establish from the risks that we know the magnitude of crises (because we do not know all the risks). We do not know the probability distribution; we are only able to observe the outcomes.

Lastly, we have a more general problem of a philosophical nature, namely the problem of induction. Why should we believe that the past can be a reliable representation of the future? Logical answers to this problem do not exist. Our knowledge is limited to past experiences, but there is no rational reason to believe that what has already happened to us can happen to us again tomorrow, with absolute certainty. This problem lies at the base of all forecasts, whether they derive from mathematical and statistical models, or whether they derive from personal beliefs based on one's own experience. To remedy this problem, Taleb introduces the concept of the fourth quadrant. This is a tool that allows us to understand when making forecasts is possible or is not dangerous, from those situations in which making forecasts can be dangerous. Below, I report the image of the fourth quadrant taken from the article “The fourth quadrant: a map of the limits of statistics” by Nassim Nicholas Taleb [09/14/2008].

APPLICATION	Simple payoffs	Complex payoffs
DOMAIN		
Distribution 1 (“thin tailed”)	Extremely robust to Black Swans	Quite robust to Black Swans
Distribution 2 (“heavy” and/or unknown tails, no or unknown characteristic scale)	Quite robust to Black Swans	LIMITS of Statistics – extreme fragility to Black Swans

Figure 7. The fourth quadrant

The map of the fourth quadrant is articulated on two axes: the rows represent the probability distributions of the phenomenon that we are analysing. We can simplify the discussion by dividing

the distributions into two cases: Gaussian phenomena and non-Gaussian phenomena. The columns, instead, represent the payoffs, and here too we can distinguish two categories: simple (binary) and complex (non-binary). This map should orient our decisions as a function of the complexity of a phenomenon.

The first and the second quadrants represent *Mediocristan*; in the first, the outcomes are binary (true or false), mathematical models work well, and their application does not worry us. In the second quadrant, the outcomes are complex, but the models can still describe a phenomenon adequately. The risks are low in the second quadrant because the effects of extreme observations are contained. The third and the fourth quadrants represent *Extremistan*. In the third quadrant, the outcomes are binary; the possibility of extreme events does not affect the outcomes. This is what is observed when we buy insurance. The fourth quadrant, instead, is the “domain of black swans”; here, extreme events have a relevant weight and their impact can be enormous. Black swans can be both positive and negative. The true challenge, therefore, is how to protect oneself from the negative ones and, at the same time, derive advantage from the positive ones. Robustness and antifragility are based on these ideas, that is, on how to protect oneself and how to exploit extreme events. But of these themes, we will speak in the next chapter. In the following paragraph, instead, we will speak of modern portfolio theory, of what the assumptions are at the base of the construction of a portfolio, and we will conclude with a comparison between active and passive management.

Modern Portfolio Theory: Hypotheses, Limits, and Empirical Evidence

The models and theories at the foundation of portfolio management have as their objective an efficient management of the available capital. We define efficiency in portfolio management in terms of risk and return: the purpose is to generate the highest possible return with the lowest level of risk acceptable by the investor. The creation of a suitable portfolio for the investor therefore passes through various steps: in the first place, it is necessary to define the risk profile; following this comes the phase of asset allocation; and lastly, the security selection. The risk profile of the investor is estimated through a survey; this serves to determine the main characteristics of the portfolio. For example, to a low risk profile, a more conservative (bond-based) portfolio will correspond. To a high one, a riskier (equity-based) portfolio will correspond. The phase of asset allocation is that which quantitatively defines the appropriate proportions for each asset class within the portfolio. Finally, the asset selection deals with the selection of the individual securities included within each asset class. In practice, after having estimated the risk profile of the investor, one proposes to them the allocation of their capital into two types of portfolios. The first portfolio is the risk-free one and the second,

instead, is the risky one. The proportions of the two portfolios vary in such a way as to make the risk-return relationship coincide with the risk profile of the investor.

The theoretical models that stand at the foundation of this process (Markowitz, SIM, CAPM, APT) are based, however, upon some important assumptions, which are worth remembering insofar as the debate on the validity of such assumptions is still open.

In the preceding paragraph, we discussed the hypothesis of Gaussian returns, verifying that empirically these do not follow a normal distribution. A Student's t-distribution would seem more accurate, but it does not eliminate the problem of fat tails. In fact, the observation of extreme returns renders questionable the assumption of the standard deviation as an adequate measure to represent the risk of an instrument (and more generally of a portfolio). Furthermore, the standard deviation does not distinguish between positive and negative variations of returns, penalizing losses and gains in the same way. This assumption, however, characterizes all the processes of portfolio construction, and to invalidate it completely would lead to invalidating all the models of portfolio optimization. Therefore, what one does is to assume that returns are normal and treat extreme observations as rare exceptions. One attempts to compensate for this defect by measuring risk in different ways: in addition to standard deviation, measures such as VaR, expected shortfall, or the Sortino ratio are utilized.

The assumption of rational expectations, instead, characterizes the choices of the investor. This allows us to model their choices, assuming for example that, all other returns being equal, they choose the portfolio with lower risk. In reality, we know that the investor is not always rational and their decisions are influenced by cognitive biases and emotional conditions.

"It is more probable that investors will continue to invest if you tell them that, on average, they will lose all of their money every thirty years, rather than if you tell them that they have a 3.3 percent probability of losing a certain sum every year"¹¹

Furthermore, we can have entirely rational investors who assign different probabilities to the same event. Current techniques for the creation of portfolios and the formulation of expectations do not include subjective expectations.

¹¹ Taleb N.N, Robustezza e Fragilità (2020), Il Saggiatore, Milano p. 98

The hypothesis of efficient markets is based on the assumption that market prices represent the available information. When a new piece of information is made public, its effect is substantiated in a movement in the price. By applying this concept, it is possible to measure the speed with which the market absorbs new information. A study conducted by Patell and Wolfson has shown how prices adjust themselves in a very fast manner to the announcement of dividends. It is common to distinguish between three forms of efficiency: weak, semi-strong, and strong. Efficiency in weak form implies that prices already reflect the information present in past prices. Consequently, it is not possible to make profits through technical analysis. Semi-strong efficiency provides that prices contain past information and the information contained in financial statements. This level implies that even through fundamental analysis, one cannot produce extra-profits in a systematic manner. Efficiency in strong form, instead, provides that prices reflect all information, even that relating to insiders. Empirical evidence demonstrates that insiders are still capable of making extra profits. Therefore, prices do not contain this information. The most credited hypothesis is that markets are efficient in semi-strong form. Like the other hypotheses, this one also has its limits, and although it would seem to be in part empirically verified, it is not capable of explaining certain phenomena such as speculative bubbles, market anomalies, and behavioural finance.

In the context of wealth management, the hypothesis of efficient markets undermines the validity of active portfolios. The concept is the following: if markets are efficient in semi-strong form, then it is not possible to derive any advantage from the selection of individual securities because the analyses of managers do not contain new information. Consequently, it will be nearly impossible for an active fund to beat the market systematically. For this reason, the most efficient choice would be to purchase products that replicate the trend of the market in general. This fact would seem to be demonstrated empirically: active funds are not capable of outperforming the reference benchmark systematically. This applies especially if we examine performances net of the fees that the investor pays. From the SPIVA¹² report of the end of 2024, of which I report the most significant table, the following picture emerges.

¹² SPIVA U.S. Scorecard 2024 (https://www.spglobal.com/spdji/en/documents/spiva/spiva-us-year-end-2024.pdf?utm_source=chatgpt.com)

**Report 1b: Percentage of U.S. Equity Funds Underperforming Their Benchmarks
(Based on Risk-Adjusted Return)**

Fund Category	Comparison Index	3-Year (%)	5-Year (%)	10-Year (%)	15-Year (%)	20-Year (%)
All Domestic Funds	S&P Composite 1500	88.96	90.04	93.36	97.99	97.26
All Large-Cap Funds	S&P 500	84.96	86.21	91.54	98.13	95.48
All Mid-Cap Funds	S&P MidCap 400	71.01	75.00	76.50	84.75	90.68
All Small-Cap Funds	S&P SmallCap 600	54.40	57.78	79.32	87.89	89.08
All Multi-Cap Funds	S&P Composite 1500	88.25	88.15	92.40	96.42	96.63
Large-Cap Growth Funds	S&P 500 Growth	64.26	83.33	92.78	99.18	100.00
Large-Cap Core Funds	S&P 500	76.11	79.45	94.77	98.40	96.33
Large-Cap Value Funds	S&P 500 Value	88.45	77.32	88.20	90.77	83.26
Mid-Cap Growth Funds	S&P MidCap 400 Growth	85.12	75.00	72.67	85.71	89.89
Mid-Cap Core Funds	S&P MidCap 400	58.82	74.07	87.20	86.27	94.31
Mid-Cap Value Funds	S&P MidCap 400 Value	62.86	60.71	76.81	79.41	83.72
Small-Cap Growth Funds	S&P SmallCap 600 Growth	68.14	63.68	81.28	90.72	93.97
Small-Cap Core Funds	S&P SmallCap 600	41.25	56.87	81.92	87.82	88.50
Small-Cap Value Funds	S&P SmallCap 600 Value	29.85	41.56	72.95	80.88	85.57
Multi-Cap Growth Funds	S&P Composite 1500 Growth	78.09	86.91	93.00	97.24	96.88
Multi-Cap Core Funds	S&P Composite 1500	85.78	89.04	97.59	99.21	96.76
Multi-Cap Value Funds	S&P Composite 1500 Value	89.05	89.03	92.11	93.75	87.50
Real Estate Funds	S&P United States REIT	-	76.83	72.53	81.00	86.42

Source: S&P Dow Jones Indices LLC, CRSP. Data as of Dec. 31, 2024. Past performance is no guarantee of future results. Table is provided for illustrative purposes.

Figure 8. SPIVA Report 2024

the most important datum is that net of fees, after a period of 20 years, almost no equity fund has beaten the benchmark. A similar situation exists also for European UCITS funds, considered to be of high quality. According to what is reported by ESMA¹³, active funds do not beat the relative passive ETF. Furthermore, net of inflation, mixed and bond funds record negative real returns.

Table 3: Comparison between the performances of active funds and European ETFs

UCITS	Funds (non-ETF)			ETFs
	Equity	Bond	Mixed	Equity
Costs and performance (2019–2023)				
Costs (% , per annum (p.a.))	2.1	1.4	2.0	0.6
Ongoing charges	1.47	0.94	1.48	0.23
Subscription fees	0.56	0.45	0.44	0.21

¹³ ESMA Market Report on Costs and Performance of EU Retail Investment Products 2024, p.6

Redemption fees	0.04	0.05	0.06	0.16
Net performance (% , p.a.)	5.8	-0.4	1.4	8.2
Change in ongoing costs (% , 2019–2023)	-4.9	-13.0	-2.1	-26.7
Inflation (% , p.a.)	4.1	4.1	4.1	4.1
Net real performance (% , p.a.)	1.7	-3.7	-2.7	4.1

These reports demonstrate that even institutional investors are not capable of beating the market, in support of the hypothesis of a semi-strong efficiency of the markets. Obviously, exceptions exist, but these concern individual managers with extraordinary capacities, such as, for example, Warren Buffett or George Soros.

Conclusions

In this chapter, we have retraced the principal historical events and the mathematical discoveries that marked the birth of statistics and econometrics, pausing on the innovative contribution brought by econophysics in the interpretation of financial markets. Through the metaphors of Mediocristan and Extremistan, we have illustrated the fundamental limits of the bell curve, showing how the normal distribution tends to underestimate the probability and the impact of extreme events, while the approach based on power laws demonstrates itself, in these contexts, decidedly more effective.

The empirical analysis of financial returns has brought to light the inadequacy of the Gaussian distribution in representing the complexity and the unpredictable nature of the markets, underlining the necessity of revising the methodologies of risk management. In this perspective, the notion of the fourth quadrant was introduced, useful for distinguishing between scenarios in which it is permissible to adopt linear models with caution and situations in which such instruments lead to grave forecasting errors. The fourth quadrant functions as a navigation map of complexity and suggests prudential strategies to minimize exposure to catastrophic risks, favouring the transition toward the third quadrant.

Finally, the core of modern portfolio theory was presented and empirical data were discussed which show how even institutional investors, despite competencies and resources, rarely manage to achieve returns superior to the market in the long period. This observation invites one to call into question the classical assumptions and opens the way to the exploration of new paradigms of robustness and antifragility in the management of financial risk, themes that will be the object of in-depth analysis in the successive chapters.

2. Fragility, Robustness, and Antifragility

In this chapter, we will introduce the formal definitions that will allow us to understand to what extent and in what way the phenomenon we are analysing is exposed to the dangers or to the opportunities of the fourth quadrant. We begin, therefore, with a simple example to introduce in an intuitive manner the definitions of fragile, antifragile, and robust. Subsequently, we will analyse the mathematical definitions proposed in the article “Taleb, N. N., & Douady, R. (2014). Mathematical definition, mapping, and detection of (anti)fragility”. Following this, we will examine some of the risk measures and assessment tools commonly used in practice. Finally, we will focus on the financial tools and strategies that permit the reduction of fragility within a portfolio.

A fragile object is such that, if subjected to a shock, it breaks easily. A term to express the opposite of fragile does not exist, for which reason Taleb has coined the term *antifragile* to indicate an object that, subjected to a shock, does not break but, on the contrary, improves. Let us take for example a ceramic cup: how can we measure the risk that a cup placed on a table might break? Clearly, if nothing happens and no one touches it, it cannot break by itself. To calculate the risk that it falls, we must therefore formulate hypotheses regarding the events that could cause a displacement of the cup such as to make it fall and break. These events are indeterminate and unpredictable; therefore, even formulating hypotheses, we are not able to determine with certainty the probability that the cup falls or that it breaks because of some perturbation.

And yet, we know that the cup is a fragile object, and it is so for a reason that we can measure. If we let the cup fall from one millimetre of height, it does not break. Conducting various attempts, it will be possible to identify a minimum height beyond which the cup, falling, breaks. Or we could hit it with different levels of force to understand with what intensity it is possible to break it. Fragility can be measured even in the absence of a known probability distribution; risk, on the contrary, requires the knowledge or the estimation of the probability of events. Measuring fragility constitutes a possible solution to the problem of the *Black Swan*, that is, the impossibility of calculating the risk that rare events of great impact occur and of predicting their occurrence. It is easier to manage the sensitivity to damages caused by volatility than to predict the event that could cause that damage. It is possible to move from the fragile to the antifragile by reducing fragility or by exploiting antifragility. Moving

in this sense means passing from the fourth quadrant to the third. Exploiting antifragility, instead, means remaining in the fourth quadrant by exposing oneself only to "positive fragility," which is to say, to antifragility.

Identifying fragility or antifragility is almost always possible through a simple test: anything that derives advantage from shocks or random events is antifragile; the contrary is fragile. Robustness, conceptually, is nothing other than a fragility contained below a determined level considered acceptable. We now deepen the theme from a more technical point of view.

Mathematical definitions of Fragility, Robustness, and Antifragility

The tail Vega sensitivity is the measure that allows us to establish when a system is fragile. This measure depends on the absolute semi-deviation (s) of the distribution, below a certain level Ω . The semi-deviations are defined in the norm L^1 (that is, in terms of the expected value of the absolute deviations) to ensure the existence of the measure even in the presence of fat-tailed distributions.

Given a random variable X , belonging to a family of probability density functions (pdf) f_λ with a single parameter, $\lambda \in I \subset \mathbb{R}$, and a fixed reference value Ω , the absolute left semi-deviation of X from Ω is defined as:

$$\text{Eq. 9} \quad s^-(\lambda) = \int_{-\infty}^{\Omega} (\Omega - x) f_\lambda(x) dx$$

We assume that $\lambda \rightarrow s^-(\lambda)$ is continuous, strictly increasing in the interval $\mathbb{R}_+ = [0, \infty)$, such that it is possible to use the left semi-deviation s^- as the parameter of the inverse function $\lambda(s): \mathbb{R} \rightarrow I$, defined by $s^-(\lambda(s)) = s$ for $s \in \mathbb{R}_+$.

This condition is satisfied if for every $x < \Omega$, the probability function is continuous and increasing in λ . In fact, the cumulative distribution function (CDF) is given by:

$$\text{Eq. 10} \quad F_\lambda(x) = Pr_{f_\lambda}(X < x) = \int_{-\infty}^x f_\lambda(t) dt$$

We obtain:

$$\text{Eq. 11} \quad s^-(\lambda) = \int_{-\infty}^{\Omega} F_{\lambda}(x) dx$$

This occurs, for example, when λ is a scale parameter, that is $x \sim \Omega + \lambda(X_1 - \Omega)$; in fact, in this case we have:

$$\text{Eq. 12} \quad F_{\lambda}(x) = F_1\left(\Omega + \frac{x - \Omega}{\lambda}\right), \quad \frac{\partial F_{\lambda}}{\partial \lambda}(x) = \frac{\Omega - x}{\lambda} f_{\lambda}(x)$$

and

$$s^-(\lambda) = \lambda s^-(1)$$

We observe, therefore, that when λ is a scale parameter, s^- is directly proportional to λ . When instead λ is a shift parameter, that is $X \sim X_0 - \lambda$, then $F_{\lambda}(x) = F_0(x + \lambda)$ and $\frac{\partial s^-}{\partial \lambda} = F_{\lambda}(\Omega)$.

Now we take a value $K < \Omega$: K represents a fixed stress level. For $s \in \mathbb{R}_+$ we define:

$$\text{Eq. 13} \quad \xi(K, s^-) = \int_{-\infty}^K (\Omega - x) f_{\lambda(s^-)}(x) dx$$

Therefore, if $K = \Omega$ we have that $\xi(\Omega, s^-) = s^-$. Assuming that the function $\xi(K, s^-)$ is differentiable in $((-\infty, \Omega] \times \mathbb{R}_+)$, the left-tail Vega sensitivity of X at the stress level $K < \Omega$ at the deviation level $s^- > 0$ for the pdf f_{λ} is:

$$\text{Eq. 14} \quad V(X, f_{\lambda}, K, s^-) = \frac{\partial \xi}{\partial s^-}(K, s^-) = \left(\int_{-\infty}^K (\Omega - x) \frac{\partial f_{\lambda}}{\partial \lambda}(x) dx \right) \left(\frac{ds^-}{d\lambda} \right)^{-1}$$

In terms of finite differences:

$$\begin{aligned} \text{Eq. 15} \quad V(X, f_{\lambda}, K, s^-, \Delta s) &= \frac{1}{2\Delta s} (\xi(K, s^- + \Delta s) - \xi(K, s^- - \Delta s)) \\ &= \int_{-\infty}^K (\Omega - x) \frac{f_{\lambda(s^- + \Delta s)}(x) - f_{\lambda(s^- - \Delta s)}(x)}{2\Delta s} dx \end{aligned}$$

Note that:

$$\text{Eq. 16} \quad \xi(K, s^-) = -E_{f_\lambda}[X|X < K]Pr_{f_\lambda}(X < K)$$

$\xi(K, s^-)$ can be decomposed into two parts:

$$\text{Eq. 17} \quad \xi(K, s^-) = (\Omega - K)F_\lambda(K) + P_\lambda(K)$$

With:

$$\text{Eq. 18} \quad P_\lambda(K) = \int_{-\infty}^K (K - x)f_\lambda(x)dx$$

The first, $(\Omega - K)F_\lambda(K)$, is proportional to the probability of the variable X being below the stress level K . The second, $P_\lambda(K)$, represents the expectation of the amount by which is below X is below K .

Letting $\lambda = \lambda(s^-)$ and integrating, one obtains:

$$\text{Eq. 19} \quad \xi(K, s^-) = (\Omega - K)F_\lambda(K) + \int_{-\infty}^K F_\lambda(x)dx = \int_{-\infty}^\Omega F_\lambda^K(x)dx$$

Where $F_\lambda^K = F_\lambda(\min(x, K)) = \min(F_\lambda(x), F_\lambda(K))$ such that:

$$\text{Eq. 20} \quad V(X, f_\lambda, K, s^-) = \frac{\partial \xi}{\partial s^-}(K, s^-) = \frac{\int_{-\infty}^\Omega \frac{\partial F_\lambda^K}{\partial \lambda}(x)dx}{\int_{-\infty}^\Omega \frac{\partial F_\lambda}{\partial \lambda}(x)dx}$$

In terms of finite differences, however:

$$\text{Eq. 21} \quad V(X, f_\lambda, K, s^-, \Delta s) = \frac{1}{2\Delta s} \int_{-\infty}^\Omega (\Delta F_{\lambda, \Delta s}^K(x)) dx$$

Where λ_s^+ and λ_s^- are such that:

$$s(\lambda_s^+) = s^- + \Delta s, \quad s(\lambda_s^-) = s^- - \Delta s$$

$$\Delta F_{\lambda, \Delta s}^K(x) = F_{\lambda_s^+}^K(x) - F_{\lambda_s^-}^K(x)$$

The function $\xi(K, s^-)$ therefore represents the expected damage below a certain stress level K . The first term, $(\Omega - K)F_\lambda(K)$, expresses the expected loss as a function of the probability of the extreme event; the second represents the average loss upon exceeding the threshold K .

$$\text{Eq. 22} \quad \xi(K, s^-) = \int_{-\infty}^{\Omega} F_\lambda^K(x) dx$$

shows that it is possible to calculate the damage through the integral of the function F_λ^K truncated at the threshold K . The Vega is $V(X, f_\lambda, K, s^-)$ the ratio between two sensitivities: in the numerator we have the sensitivity to the damage below K , while in the denominator we have the sensitivity to the left semi-deviation (that is, the damage of the non-truncated tail).

Definition of Fragility

Now that we have defined the measure of Vega sensitivity, we can mathematically define fragility. Briefly, fragility is the sensitivity of a given risk measure to an error in the deviation parameter of a distribution.

Intrinsic Fragility

The local fragility of a random variable X_λ depending on the parameter λ at the stress level K and at the semi-deviation level $s^-(\lambda)$ with the pdf f_λ , is its Vega $V(X, f_\lambda, K, s^-)$.

The fragility expressed in terms of finite differences of a random variable X_λ depending on the parameter λ at the stress level K and at the semi-deviation level $s^-(\lambda) + \Delta s$ with the pdf f_λ , is its $V(X, f_\lambda, K, s^-, \Delta s)$.

In this definition, the fragility resides in the implicit assumption made in the extrapolation of the distribution of X_λ , from the areas used to estimate $s^-(\lambda)$ around Ω , to the areas around K on which the risk measure depends ξ .

Inherited Fragility

Let us consider the particular case in which a random variable $Y = \varphi(X)$ depends on a source of risk X , itself subject to a parameter λ . We maintain unchanged the notation used for X , while we denote with g_λ the pdf of Y , $\Omega_Y = \varphi(\Omega)$ and with $u^-(\lambda)$ the left semi-deviation of Y . Given a certain level $L = \varphi(K)$, we define as in the case of X :

$$\text{Eq. 23} \quad \xi(L, u^-(\lambda)) = \int_{-\infty}^K (\Omega_Y - Y) g_\lambda(y) dy$$

The inherited fragility of Y with respect to X at the stress level $L = \varphi(K)$ and at the semi-deviation $s^-(\lambda)$ of X , is the partial derivative:

$$\text{Eq. 24} \quad V_X(Y, g_\lambda, L, s^-(\lambda)) = \frac{\partial \xi}{\partial s^-}(L, u^-(\lambda)) = \left(\int_{-\infty}^K (\Omega_Y - y) \frac{\partial g_\lambda}{\partial \lambda}(y) dy \right) \left(\frac{ds^-}{d\lambda} \right)^{-1}$$

The stress level and the probability density function are defined for the variable Y , while the parameter utilized for the differentiation is the absolute left semi-deviation of X , $s^-(\lambda)$. In fact, in this process, one first measures the distribution of X and its absolute left semi-deviation. Subsequently, one applies the function φ , utilizing a mathematical model of Y with respect to X , and one estimates the risk measure ξ . If an error is committed in measuring $s^-(\lambda)$, its impact on the risk measure of Y is amplified by the ratio given by the "inherited fragility."

From a graphical point of view, we can distinguish two types of fragility. The first type is typical of financial returns, the distribution of which has heavy tails both to the right and to the left. The second type of fragility is instead characterized by a heavy left tail and an absent right tail.

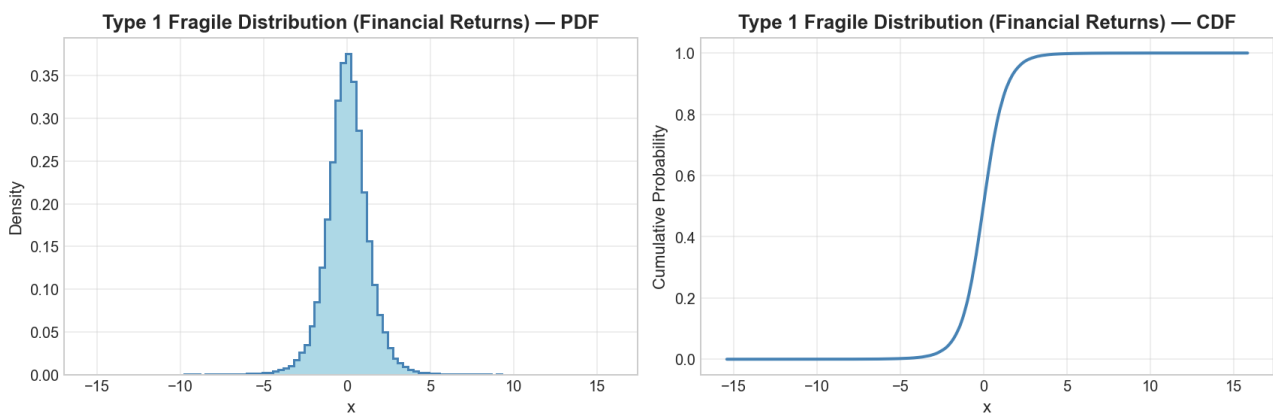


Figure 9. Type one fragility

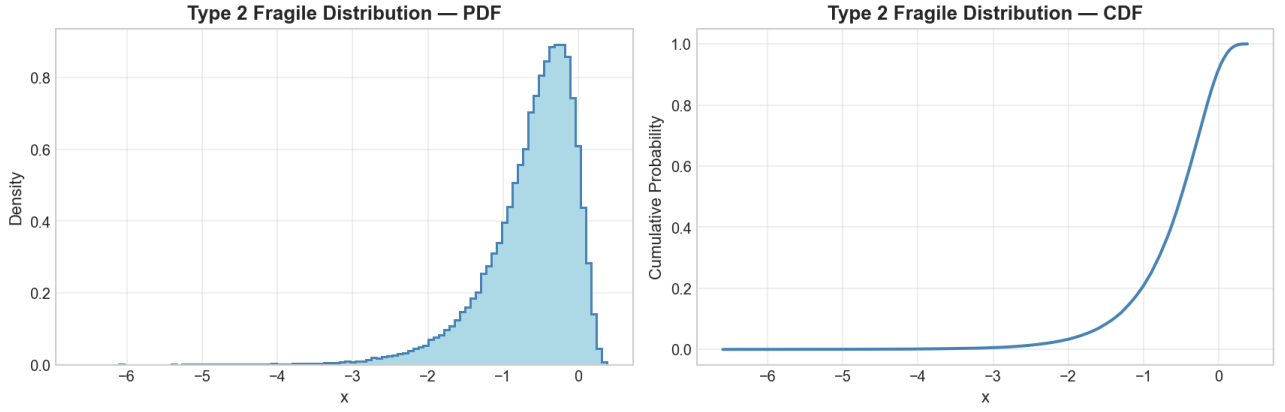


Figure 10. Type two fragility

We now define two functions that allow us to further investigate the dynamics of the fragility of a system: the first is the fragility drift and the second is the second-order fragility.

The Fragility Drift

We define the Fragility drift as:

$$\text{Eq. 25} \quad V'_K(X, f_\lambda, K, s^-) = \frac{\partial^2 \xi}{\partial K \partial s^-}(K, s^-)$$

Fragility always manifests because of other fragility; in fact, by definition we know that $\xi(\Omega, s^-) = s^-$, therefore $V(X, f_\lambda, \Omega, s^-) = 1$. The fragility drift measures the speed with which fragility separates from its original value 1 as K moves away from the centre Ω .

Second-order Fragility

The second-order fragility is the second derivative of the estimated tail $\xi(K, s^-)$ with respect to the absolute semi-deviation s^- :

$$\text{Eq. 26} \quad V'_{s^-}(X, f_\lambda, K, s^-) = \frac{\partial^2 \xi}{(\partial s^-)^2}(K, s^-)$$

The second-order fragility drives the bias in the estimates of the stress test when the value of s^- is subject to uncertainty, through Jensen's inequality.

Fragility Transfer Theorem

We need an instrument that allows us to understand when an object is more fragile than another. The Fragility Transfer Theorem states the following:

THEOREM: Let $\varphi: \mathbb{R} \rightarrow \mathbb{R}$ be a continuous twice-differentiable function such that: $\varphi(\Omega) = \Omega$ and for every $x < \Omega$, $\frac{d\varphi}{dx}(x) > 0$.

The random variable $Y = \varphi(x)$ is more fragile at level $L = \varphi(K)$ with pdf g_λ , with respect to the variable X at level K with pdf f_λ , if and only if:

$$\text{Eq. 27} \quad \int_{-\infty}^{\Omega} H_{\lambda}^{\Omega}(x) \frac{d^2 \varphi}{dx^2}(x) dx < 0$$

where:

$$\text{Eq. 28} \quad H_{\lambda}^{\Omega}(x) = \frac{\partial P_{\lambda}^K(x)/\partial \lambda}{\partial P_{\lambda}^K(\Omega)/\partial \lambda} - \frac{\partial P_{\lambda}(x)/\partial \lambda}{\partial P_{\lambda}(\Omega)/\partial \lambda}$$

And where:

$$\text{Eq. 29} \quad P_{\lambda}(x) = \int_{-\infty}^x F_{\lambda}(t) dt, \quad P_{\lambda}^K(x) = \int_{-\infty}^x F_{\lambda}^K(t) dt$$

We can interpret $P_{\lambda}(x)$ as the price of a Put option on X_{λ} with strike price x and $P_{\lambda}^K(x)$ as a Put option with strike price x and a European “down-and-in” barrier at the price level K . H can be seen as a transfer function, expressed as the difference between two ratios. For a given level x of the random variable on the left side of Ω ; the second ratio is given by the Vega of the Put with strike x normalized by the Vega of a Put with strike ATM (that is, with strike price at Ω). While the first is the same ratio, but in this case the Put options with strike price x and Ω , are instead European Put options with a “down-and-in” barrier with trigger price K .

Therefore:

$$\text{Eq. 30} \quad V(Y, g_{\lambda}, L, u^{-}) - V(X, f_{\lambda}, K, s^{-}) = - \int_{-\infty}^{\Omega} H_{\lambda}^{\Omega}(x) \frac{d^2 \varphi}{dx^2}(x) dx$$

Where $V(Y, g_{\lambda}, L, u^{-})$ is the European option with a barrier and $V(X, f_{\lambda}, K, s^{-})$ is the classic European option.

When H is equal to zero, the system is robust: a transformation of the left tail for a threshold L less than K does not produce a decrease in fragility. When H is less than zero, then it means that by applying the transformation φ the fragility decreases; vice versa, if H is greater, the transformation amplifies the fragility. The increase in fragility is therefore attributable to concave transformations of the starting distribution.

Definition of Antifragility and Robustness

Having concluded the discussion on fragility, we can pass to the mathematical definition of the concepts of robustness and antifragility. We recall that the concept of antifragile is not the exact opposite of fragile. In fact, an antifragile system requires robustness in the left tail of the distribution and fragility in the right one.

Definition of robustness

Let us assume that X_λ belongs to a family of random variables with a single parameter and with a pdf f_λ . Robustness is an upper control of the fragility of X residing in the left side of the distribution.

We say that f_λ is b -resistant beyond the stress level $K < \Omega$ if $V(X_\lambda, f_\lambda, K', s^-(\lambda)) \leq b$ for every $K' < K$. In other words, the robustness of f_λ on the semi-line $(-\infty, K]$ is $R_{(-\infty, K]}(X_\lambda, f_\lambda, K, s^-(\lambda)) = \max_{K' < K} V(X_\lambda, f_\lambda, K', s^-(\lambda))$ so that b -resistant simply means $R_{(-\infty, K]}(X_\lambda, f_\lambda, K, s^-(\lambda)) \leq b$.

We also define the robustness in a given interval $[K_1, K_2]$ with the same inequality valid for every $K' \in [K_1, K_2]$. In this case we use: $R_{[K_1, K_2]}(X_\lambda, f_\lambda, K, s^-(\lambda)) = \max_{K_1 \leq K' \leq K_2} V(X_\lambda, f_\lambda, K', s^-(\lambda))$

Substantially, b represents the maximum level of fragility that we are willing to accept to define a system as robust. It is an arbitrarily defined threshold, which must be valid for all values of V calculated for K' lower than K . If even a single V exceeds b , then the system is not robust. R is a function of robustness, which expresses the maximum value of fragility V in the interval $(-\infty, K]$. Whereby we can say that a system is robust if the maximum encountered value of V is lower than b . We note that the lower R is, the tighter the control and the more robust is the distribution pdf f_λ .

Graphically, a robust probability distribution appears as a curve without heavy tails, that is, more similar to the Gaussian.

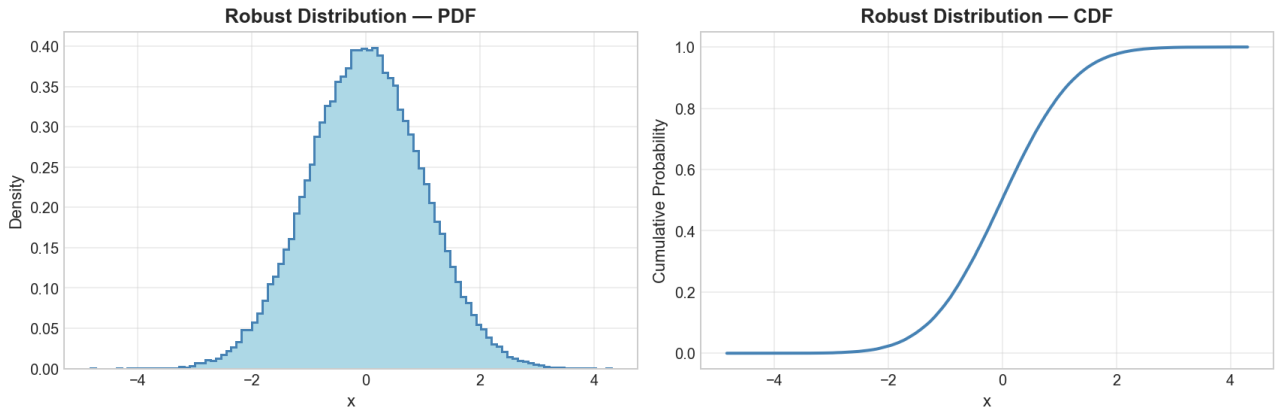


Figure 11. Robust phenomenon

Definition of Antifragility

The second condition of antifragility concerns the right side of the distribution. Let us define the right absolute semi-deviation of X :

$$\text{Eq. 31} \quad s^+(\lambda) = \int_{\Omega}^{+\infty} (x - \Omega) f_{\lambda}(x) dx$$

$s^+(\lambda)$ is a measure of the expected gain or benefit for values above the threshold Ω . The expected gain in a specific interval $[L, H]$ with $H > L > \Omega$ will be given by:

$$\text{Eq. 32} \quad \xi^+(L, H, s^+(\lambda)) = \int_L^H (x - \Omega) f_{\lambda}(x) dx$$

The measure of sensitivity to the right tail for variations of λ in the interval $[L, H]$ is:

$$\begin{aligned} \text{Eq. 33} \quad W(X, f_{\lambda}, L, H, s^+) &= \frac{\partial \xi^+(L, H, s^+)}{\partial s^+} \\ &= \left(\int_L^H (x - \Omega) \frac{\partial f_{\lambda}}{\partial \lambda}(x) dx \right) \left(\int_{\Omega}^{+\infty} (x - \Omega) \frac{\partial f_{\lambda}}{\partial \lambda}(x) dx \right)^{-1} \end{aligned}$$

When $Y = \varphi(x)$ is a variable dependent on a source of randomness X , we define:

$$\text{Eq. 34} \quad W_x(Y, g_\lambda, \varphi(L), \varphi(H), s^+) = \left(\int_{\varphi(L)}^{\varphi(H)} (y - \varphi(\Omega)) \frac{\partial g_\lambda}{\partial \lambda}(y) dy \right) \left(\int_{\Omega}^{+\infty} (x - \Omega) \frac{\partial f_\lambda}{\partial \lambda}(x) dx \right)^{-1}$$

W and W_x represent the opposites of fragility (V) and of inherited fragility (V_x). In this case, the higher the sensitivity, the more the system exploits variability in a positive sense. The antifragility of a system consists precisely in the capacity to exploit the variability of the right tail of the distribution, while controlling the sensitivity toward extreme events of the left tail.

An antifragile distribution is such if it benefits from the effects of the right tail but avoids the losses of the left tail. Whereby graphically an antifragile probability distribution appears as follows:

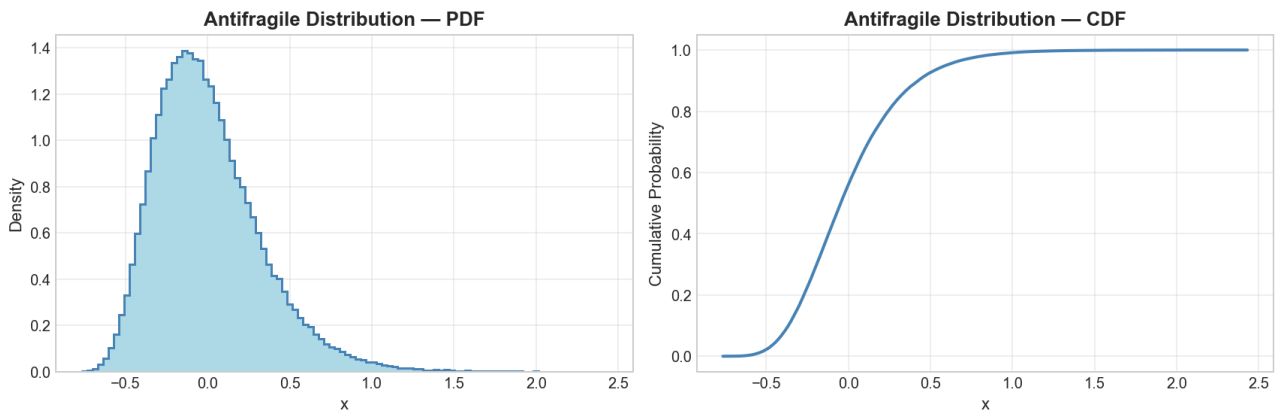


Figure 12. Antifragile phenomenon

These are the instruments proposed by Taleb for the identification and quantification of tail risk. They are concepts that can be applied to any distribution of data provided one has enough empirical data. In the following paragraph, we will take a look at traditional instruments and at some variants of instruments used in finance to evaluate risk.

Traditional instruments for the control of financial risks

The objective of this section is to introduce the traditional concepts used to measure and control the risk of financial assets and portfolios. Traditionally, the risk measure most used in finance as a proxy for risk is the volatility of returns. The volatility of a determined asset is nothing other than the standard deviation of returns. We denote with $R(T)$ the logarithmic return in the interval T , defined as:

Eq. 35
$$R(T) = \log \frac{x(T)}{x_0}$$

Where $x(T)$ is the price of asset X at time T , while x_0 is today's price. If $P(x, T|x_0, 0)dx$ is the probability of obtaining $x(T)$ in dx , the variance of returns will be defined as:

Eq. 36
$$\sigma^2 = \frac{1}{T} \left[\int P(x, T|x_0, 0) R^2(T) dx - \left(\int P(x, T|x_0, 0) R(T) dx \right)^2 \right]$$

and the volatility will therefore be $\sigma = \sqrt{\sigma^2}$. Volatility, even today, is adopted as an adequate risk measure for a given investment. This definition, however, presupposes that returns are normally distributed, that the distribution is symmetric, and that returns are independent of each other. This measure, as previously stated, assigns the same probability of occurrence to positive and negative extreme events, therefore underestimating tail risk. Given that financial risk resides above all in the events of the left tail, other measures have been developed that attempt to represent the financial risk of an instrument in a more adequate manner than volatility.

The VaR (Value at Risk) and the expected shortfall

For a better management of financial risk, it is therefore necessary to define other metrics that concentrate above all on the study of extreme events. In this regard, an interesting measure in this sense is the VaR. The VaR measures the probability of the occurrence of tail events. We define, therefore, the probability of losing an amount $-\delta x$ larger than a threshold Λ in a given time interval τ as:

Eq. 37
$$P[\delta x < -\Lambda] = P_{<}[-\Lambda] = \int_{-\infty}^{-\Lambda} P_{\tau}(\delta x) d\delta x$$

Where $P_{\tau}(\delta x)$ is the probability density of a price change in the time interval τ . Alternatively, we can define the risk as a level of loss (VaR) Λ^* corresponding to a certain probability of loss P^* in the time interval τ in the following way:

Eq. 38
$$P^* = \int_{-\infty}^{-\Lambda^*} P_{\tau}(\delta x) d\delta x$$

This is equivalent to saying, for example, that a loss higher than Λ^* , taking into consideration a daily time interval $\tau = 1$, occurs every 100 days on average for $P^* = 1\%$. This risk measure does not take into account cumulative losses; that is, it does not consider consecutive negative returns. This limitation underestimates the risk of extreme losses. By taking into consideration only single exceptional negative returns, the magnitude of a crisis is lowered. Furthermore, it does not allow us to quantify the average loss of such events. In this regard, the expected shortfall, or the CVaR (Conditional Value at Risk), was developed, defined as:

$$\text{Eq. 39} \quad E^* = \frac{\int_{-\infty}^{-\Lambda^*} P_{\tau}(\delta x) d\delta x}{P^*}$$

In this definition, both Λ^* and P^* are defined in accordance with the VaR formula.

Evaluation of portfolio performance (Sharpe Ratio and Sortino Ratio)

In relation to the assumed definition of risk and to the return, we can evaluate portfolio performances. The classic method bases the evaluation of a portfolio's performance on the Sharpe ratio, defined as:

$$\text{Eq. 40} \quad s_t = \frac{\overline{R_i} - \overline{rf}}{\sigma_i}$$

Where: $\overline{R_i}$ represents the expected return of portfolio i in the time interval τ , $\overline{rf}$ represents the average risk-free rate in τ , and σ_i is the standard deviation of returns in period τ . The Sharpe ratio is nothing other than the ratio between the average excess return of the portfolio and its volatility. This evaluation metric for portfolio performances is based on the classic notion of risk and therefore suffers from the same biases. An alternative measure is the Sortino Ratio, defined in the following way:

$$\text{Eq. 41} \quad s_t = \frac{\overline{R_i} - \Lambda}{DR_i}$$

Where: Λ is the minimum acceptable return threshold for portfolio i in period τ , while DR_i is the downside risk coefficient for the portfolio i in the considered period. The downside risk is nothing other than the semi-deviation of returns. The semi-deviation is defined in the following way:

$$\text{Eq. 42} \quad \text{semideviazione} = \sqrt{\frac{1}{n} \sum_{R < \bar{R}} (R_{ij} - \bar{R}_i)^2}$$

Where n is the number of returns lower than the expected return of the portfolio. The expected return can be replaced with 0 if the objective is to focus on negative returns. This measure approaches closer to Taleb's framework, but differs substantially due to the exponentiation. Taleb utilizes the absolute left semi-deviation as a starting point to develop his analysis on tail sensitivity.

Maximum drawdown (MDD)

The Maximum drawdown measures the worst loss that occurred in the considered period. It is a non-parametric measure that is calculated directly from market prices, not from returns.

$$\text{Eq. 43} \quad MDD = \max_{\tau \in (0, T)} D(\tau) = [\max_{t \in (0, \tau)} X(t) - X(\tau)]$$

It allows us to evaluate how risky a portfolio actually is by looking at the difference between the maximum value reached and the subsequent minimum, thus highlighting the worst percentage decline that took place in the analysed period.

Options as an instrument for the construction of robust and antifragile portfolios

Options are derivative instruments that allow the holder to exercise the right of purchase (call option) or of sale (put option) at a determined price level (strike price). Based on the definition of robustness provided previously, we can say that an instrument is robust when the sensitivity of the left tail of its returns is contained below a certain threshold deemed acceptable. In practice, this is the case for options. Options can be interpreted as bets whose loss is limited to the premium paid and whose profit is potentially unlimited. This can be observed from the payoff graphs of call and put options.

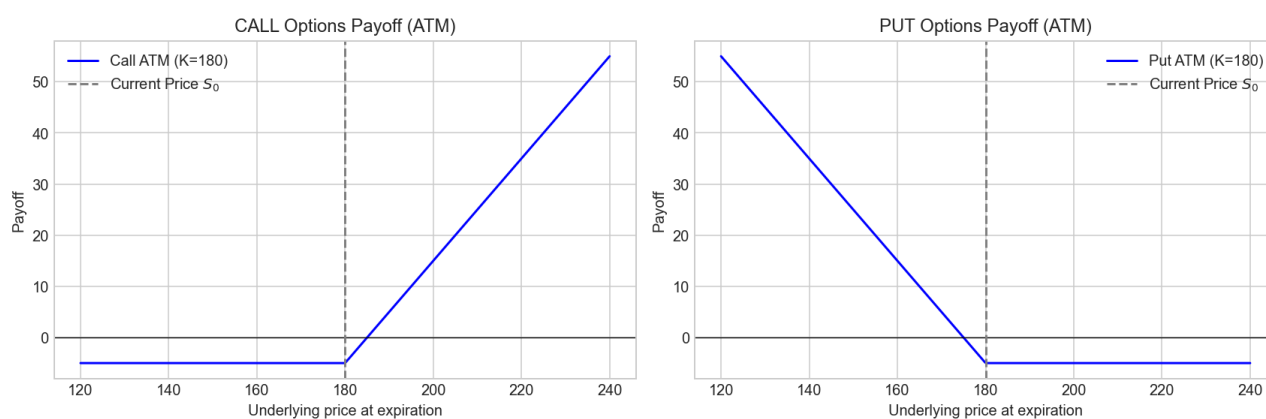


Figure 13. Call and Put options payoff

More specifically, call options are robust instruments insofar as they derive benefit from the increase in positive volatility (that is, from increases in the price of the underlying) against a bet whose loss is limited. Put options, instead, can be considered antifragile instruments, insofar as they derive benefit from the increase in negative volatility, that is, from the loss of value of the underlying asset, with a possible limited loss. This behaviour is therefore in line with that described by Taleb to define antifragility: namely, the capacity to derive profit from negative shocks. By exploiting options, it is therefore possible to construct robust or antifragile portfolios. By way of example, some strategies capable of exploiting the qualities of options are reported:

Table 4: Barbell strategies with options

Strategy	Composition	Benefit
Portfolio 1 Barbell with Call	Bond 95% + OTM long-maturity Call	The Bond guarantees a minimum level of return, covering at least in part the costs of the Call. The Call option allows one to benefit from the leverage effect at a reduced cost; the benefit will be greater if the underlying is very risky (tech stocks, small caps, crypto assets). The loss is limited to the premium paid to purchase the calls.
Portfolio 2 Barbell with Put	Bond 95% + OTM short-maturity or long-maturity American Put options	The Bond guarantees a minimum level of secure return that goes to cover at least in part the costs of the Puts. The objective is to exploit extremely negative movements in the market that occur during crises. The reduced cost of short-maturity OTM Puts allows for making numerous bets. A

further benefit is the anti-cyclical nature of such a strategy.

Portfolio 3	Bond 95% + OTM Call	Allows for exploiting both characteristics of the previous portfolios. It has the advantage of not being directional and the disadvantage of being twice as expensive.
Barbell with Put and Call	and Put on the same asset	
Portfolio 4	Index or stocks + OTM	Allows for fixing the maximum bearable loss at a predetermined level deemed compatible with the risk level of the investor. A better risk control strategy compared to simple passive investment based only on portfolio diversification. It requires the sacrifice of a part of the return to purchase the protection.
Underlying Put	with short-maturity Put	

Conclusions

In this chapter, the definitions proposed by Taleb of fragility, robustness, and antifragility have been examined. This framework offers a different perspective for risk evaluation, as it does not pose hypotheses on the data-generating model nor on the relative probability distribution. The limit of such measures is the necessity of a sufficient quantity of data; the more extreme data are available, the more accurate the estimation of the left tail sensitivity will be. Another problem of this methodology lies in the definition of the thresholds Ω and K . In the case of financial investments, we could fix Ω at 0 or equal to the mean of the returns. The threshold K , instead, represents the level starting from which we consider returns extreme: it can be subjective or calculated on the basis of the distribution. Similarly to VaR, we can establish the threshold K based on the lower quantile of the distribution of returns. However, establishing K with a data-driven methodology introduces a problem of comparability of results between different distributions. Subsequently, some of the most common metrics used to evaluate the risk and investments of financial instruments were analysed. Finally, the case of options was treated, financial instruments that allow for moving from the fourth to the third quadrant (insuring oneself against extreme losses), or for constructing portfolios capable of exploiting positive black swans, thus remaining in the fourth quadrant without being exposed to extreme negative events. In the next chapter, an empirical application of the fragility measures will be presented, examining the case of different investment portfolios. In particular, the case of a passive investment in an index will be analysed, the way in which the purchase of put options modifies the risk, and the results of different Barbell portfolios will be compared.

3. Empirical analysis on investment portfolios

In the previous chapter, we discussed the concepts of fragility, robustness, and antifragility in a theoretical manner. In this chapter, we will examine a practical application of the instruments proposed by Taleb, analysing various types of portfolios constructed by applying different strategies. Specifically, we will examine the case of passive investment in a stock index. On the same passive portfolio, we will simulate the purchase of a Put option and the effect that the latter has on the portfolio returns, comparing the results between pure passive investment and a portfolio hedged against tail risks. Finally, we will analyse the Barbell strategy by constructing different types of portfolios with different proportions in safe and risky assets. The objective is to highlight the difference of the various strategies in managing tail risk, specifically regarding their Vega sensitivity toward the left tail. Let us now examine the three strategies taken into consideration.

The passive investment strategy

This is the simplest investment strategy that an inexperienced investor can adopt. It consists of purchasing financial products that replicate the trend of the market as a whole. In the example we will examine in this chapter, we will make a simplification: that is, we will use the S&P 500 index as a proxy for the market. This strategy has two main advantages, which are simplicity and low costs. It does not require the study or analysis of markets or companies and does not require a constant updating of the portfolio. The passive strategy is based on the hypothesis that the market grows in the long period. This hypothesis is currently supported by historical facts and data, but it does not safeguard the investor from possible extreme events such as economic crises or speculative bubbles, as already demonstrated in chapter one.

The protection strategy through the purchase of Put options

The hedging of the portfolio allows for the elimination of the risk of extreme losses. Every time a crisis occurs, the Put enters into action and compensates for the negative return of the portfolio, bringing it to the minimum level acceptable to the investor. To function correctly, such a strategy needs the continuous purchase of Put options on the market. The price of Put options varies based on the underlying, the strike price, the expiration of the option, and the volatility of the market. The costs, therefore, will render the hedged portfolio less remunerative than the non-hedged one. We find the benefit in the control of tail risks. The risk/return trade-off is represented by the cost of protection. The more the portfolio is hedged, the higher the cost of the Put options.

The Barbell Strategy

The barbell strategy, proposed by Taleb, provides for the creation of an investment portfolio composed, for a share between 85% and 95%, of very low-risk assets, and for the remaining part of highly risky assets. The objective is to balance the variability of the more volatile returns with the stability of securities with contained risk. The intuition at the base of this strategy consists in exploiting the potentially rapid growth of the riskier assets, while simultaneously limiting losses thanks to the reduced share of capital invested in such instruments. In order for the strategy to function, it is necessary that the investment in the risky component be sufficiently diversified, so as to derive benefit from the high number of independent bets. This strategy represents a compromise between the two preceding ones: a very significant part of the capital is invested in instruments with very low risk (e.g., T-bills), which do not require constant monitoring but only a periodic repurchase, while the risky component is destined for instruments for which strong future growth is foreseen. In the worst-case scenario that is, if all the companies included in the risky component fail, the maximum loss would be limited to 15%, 10%, or 5%, based on the preferences of the investor.

Having terminated this brief introduction, we can proceed with the analyses. We will examine jointly the case of the passive portfolio and of the hedged one, given that the underlying of the Put coincides with the market index. Subsequently, we will analyse the Barbell portfolios and will conclude the discussion by commenting on the results of the three strategies.

All the historical series of prices utilized in the present work have been obtained through the Yahoo Finance API for Python. For simplicity, in all the analyses, transaction costs have not been considered.

Analysis of the passive portfolio with and without protection

The analysis was conducted on the weekly returns of the S&P 500. The analysed period goes from 1950 to 2025. As previously stated, phenomena belonging to Extremistan take a long time to reveal their real properties. For this reason, such an extended period of time was considered. The objective is to capture as many extreme phenomena as possible; for this reason, there are no grounds for considering short periods of time if many data are available. In this way, the real properties of the distribution of returns of the analysed phenomenon emerge.

Regarding the protected portfolio, historical prices for every strike price of the Put were not analysed. To simplify the analysis and make it replicable for other underlying assets as well, the effect that the

purchase of the Put would have had on the portfolio returns was simulated. Specifically, a strike ratio was fixed, which allows us to modulate the level of protection of the portfolio. If the strike ratio is equal to 1, the protection is complete that is, the Put options are bought At-The-Money. In the following analysis, the strike ratio was fixed at 0.97, assuming the purchase of OTM Puts. The cost of the Put was fixed based on a premium rate; in the case under examination, this is equal to 0.0002. Such a low value is based on the hypothesis that the Puts are purchased almost at expiration and OTM; specifically, the assumed expiration is the weekly one. In this way, the minimum weekly return of the portfolio was fixed. Subsequently, the payoff of the Put and the value of the portfolio were calculated as:

$$payoff\ portfolio(t) = price(t) + payoff\ put(t) - cost\ put(t)$$

Where:

$$payoff\ put(t) = \max(strike(t) - price(t), 0), \quad strike(t) = price(t) \times strike\ ratio$$

$$cost\ put(t) = price(t) \times premium\ ratio$$

From this historical series, the logarithmic returns of the hedged portfolio were calculated. Let us then analyse from a graphical point of view how the empirical distribution of the returns of these two portfolios appears.

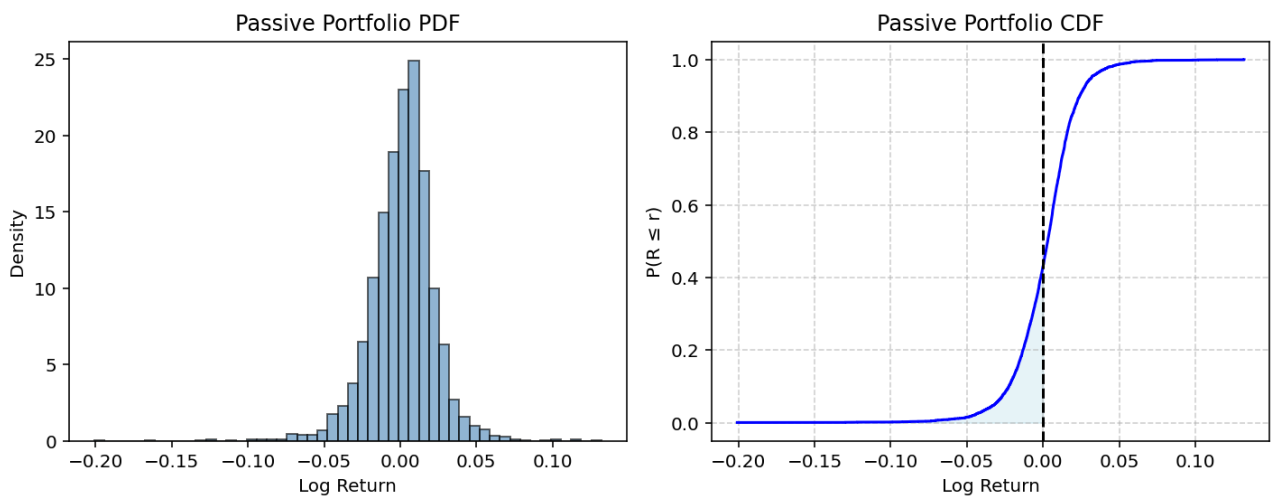


Figure 14. PDF and CDF of the passive portfolio

The distribution of the non-protected portfolio appears like the one described by Taleb for type 1 fragile phenomena, which corresponds to financial data. Specifically, we observe that the distribution is not symmetric: it is slightly shifted toward the right with a slightly positive mean. We furthermore note that the left tail is longer than the right one. This means that the observed negative returns are more extreme than the positive ones.

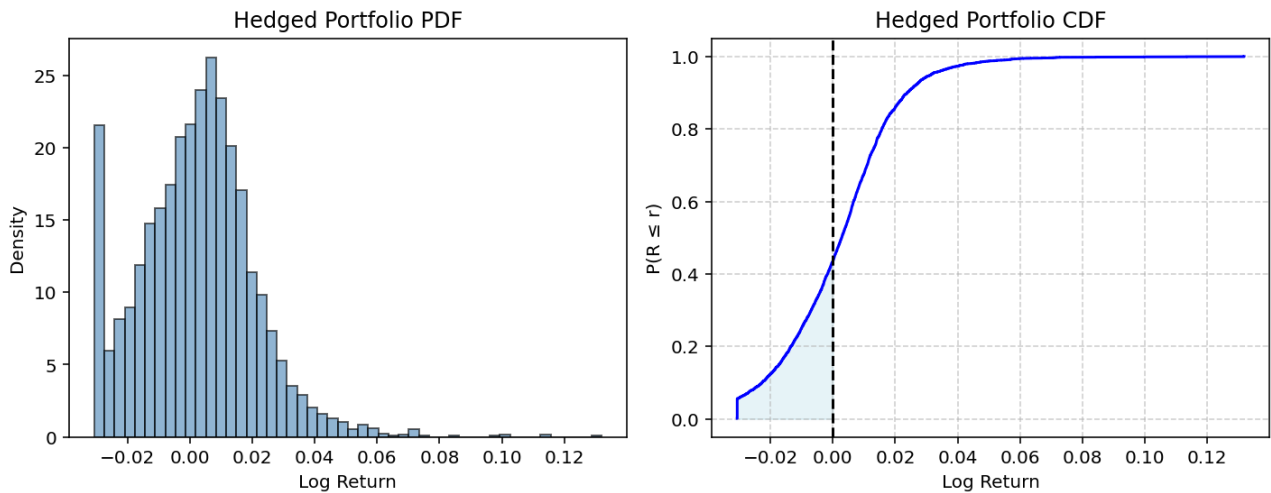


Figure 15. PDF and CDF of the hedged portfolio

The distribution of the hedged portfolio, instead, is much more similar to the one that in chapter two describes antifragile phenomena. In fact, the losses of the left tail are limited by the Put, while the right tail is not limited. The cost of the Put, however, negatively influences the right tail by lowering, albeit slightly, the positive returns.

The areas highlighted in light blue in the CDF graphs represent the absolute left semi-deviation s^- .

By subjecting the two portfolios to a stress test and assuming that a series of extreme returns occurs, we can observe how the probability and cumulative probability distributions vary.

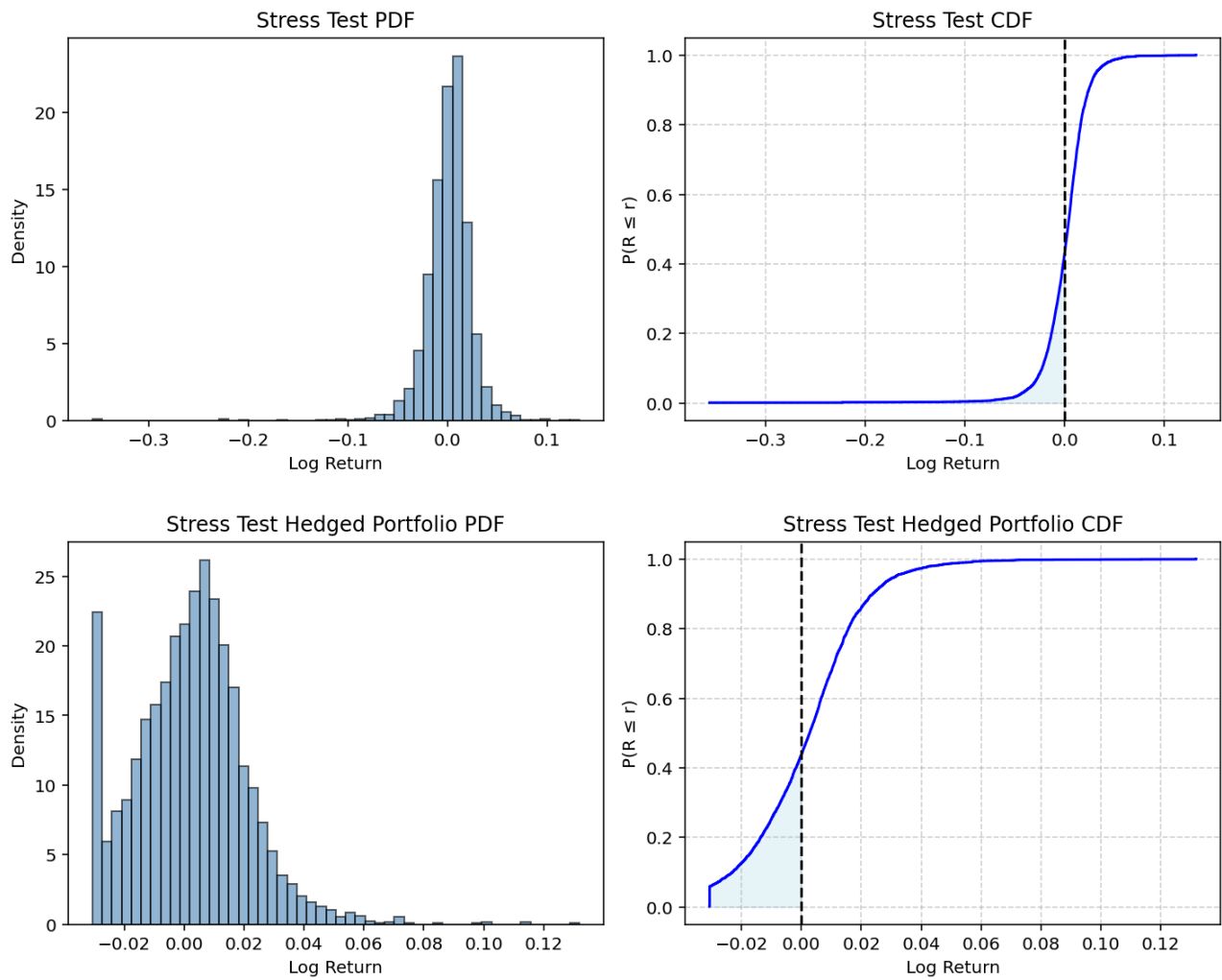


Figure 16. Changes in the PDF and CDF after the stress test

From the stress test, it emerges that the distribution of the non-hedged portfolio shows a further lengthening of the left tail, while the distribution of the hedged portfolio remains substantially unchanged. This result is also confirmed by the analysis of the following statistics:

Table 5: Summary statistics on the distributions

Portfolios	Mean	Std. deviation	Variance	Quantile (0.05)
Passive Portfolio	0.001493	0.021153	0.000447	-0.031880
Hedged Portfolio	0.002262	0.018486	0.000342	-0.030665
Stress test on passive Portfolio	0.000817	0.025201	0.000635	-0.032903
Stress test on hedged Portfolio	0.002161	0.018547	0.000344	-0.030665

Now that we have visualized the distributions and obtained the empirical CDFs, we can apply the functions analysed in the previous chapter to study the intrinsic fragility of these portfolios. The process followed to calculate the Vega is the following:

- 1) We calculate s^- from the empirical probability function by establishing $\Omega = 0$ as the threshold.
- 2) We calculate the parameter λ as an inverse function of s^- , taking the s^- of the passive portfolio as the reference value.
- 3) We calculate $\xi(s^-, K)$ utilizing various thresholds of K , so as to be able to compare the results for the various tests.
- 4) We calculate the Vega $V(X, f_\lambda, K, s^-, \Delta s)$, with $\Delta s = 1e - 3$.

Table 6: Results for K equal to -0.02

Portfolios	s^-	λ	$\xi(s^-, K)$	$V(X, f_\lambda, K, s^-)$.
Passive portfolio	0.006964	1.000000	0.004268	1.123189
Hedged portfolio	0.006084	0.873579	0.002471	0.965300
Stress test on passive portfolio	0.007614	1.093252	0.005716	1.230398
Stress test on hedged portfolio	0.006159	0.884371	0.002623	0.969907

Table 7: Results for K equal to -0.03

Portfolios	s^-	λ	$\xi(s^-, K)$	$V(X, f_\lambda, K, s^-)$.
Passive Portfolio	0.006964	1.000000	0.002704	0.930624
Hedged Portfolio	0.006084	0.873579	0.000000	0.942947
Stress test on passive portfolio	0.007614	1.093252	0.004064	1.087345
Stress test on hedged portfolio	0.006159	0.884371	0.000000	1.025091

Table 8: Results for K equal to -0.032 (that is, the quantile (0.05) of the distribution of returns of the passive portfolio)

Portfolios	s^-	λ	$\xi(s^-, K)$	$V(X, f_\lambda, K, s^-)$.
Passive Portfolio	0.006964	1.000000	0.002490	0.836177
Hedged Portfolio	0.006084	0.873579	0.000000	0.000000
Stress test on passive portfolio	0.007614	1.093252	0.003812	1.032296
Stress test on hedged portfolio	0.006159	0.884371	0.000000	0.000000

The parameter λ represents the changes in scale in the distribution of returns with respect to the original distribution, that is, that of the protected portfolio. The purchase of a Put as a hedge causes the left semi-deviation s^- to decrease. We can interpret this change as a scale variation of the original distribution equal to $\lambda = 0.873579$, obtained as $\lambda = \frac{s_{pp}^-}{s^-pc}$. We can apply the same reasoning to the other λ . Graphically:

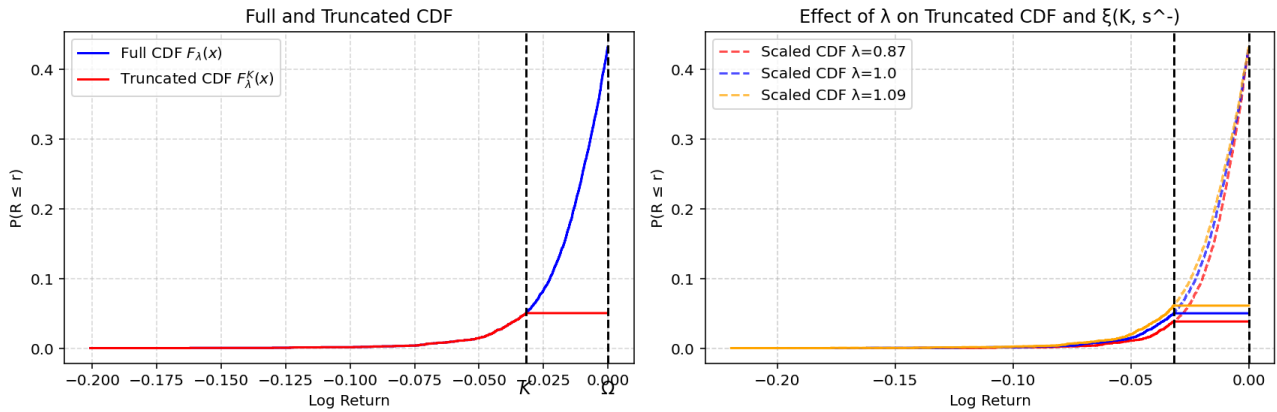


Figure 17. Effect of the scale parameter on the left side of the CDF

Let us now observe the values of $\xi(s^-, K)$ for the different levels of K . We note that as K decreases, the value ξ increases; while, as K increases, the Vega value also grows, reaching its maximum for $K \approx -0.02$. The relationship between K and ξ , and between K and Vega, is represented by the graphs that follow.

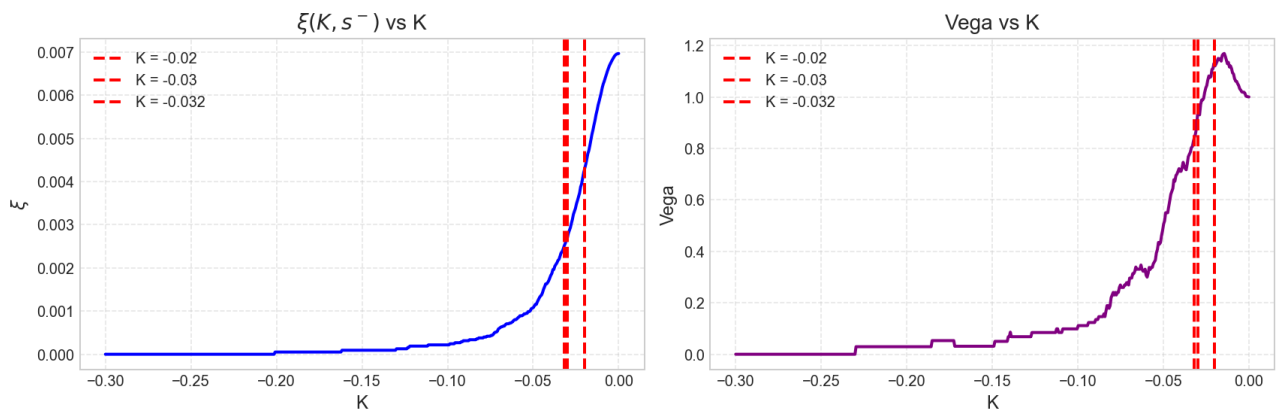


Figure 18. Passive portfolio

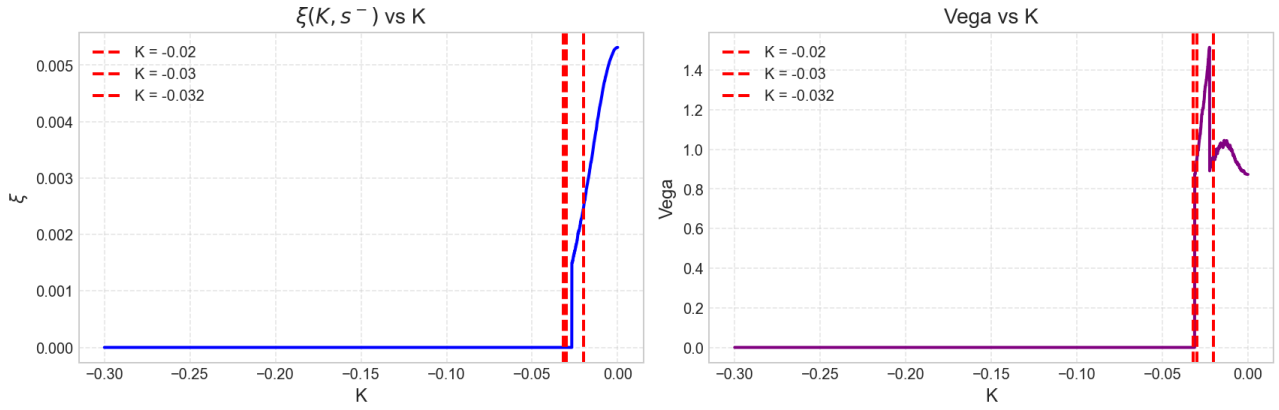


Figure 19. Hedged portfolio

The second graph highlights that, for values of K lower than the protection level of the Put, the Vega sensitivity and the value of $\xi(s^-, K)$ are reduced to 0. The Vega peak observed around the protection level offered by the Put is due to the fact that the calculation of Vega occurs through finite differences, since the empirical distribution does not allow the point-wise calculation of the derivatives of $\xi(s^-, K)$ with respect to s^- . In practice, by reducing Δs , the value of Vega around the strike price of the Put grows rapidly. This phenomenon is due principally to a problem of calculating the derivative and only to a lesser extent to an actual increase in the sensitivity of the left tail around the threshold K .

Let us now analyse how the Vega sensitivity varies as s^- varies. The stress test determined a variation of s^- in both types of portfolios (hedged and non-hedged), but the magnitude of such variation is significantly different.

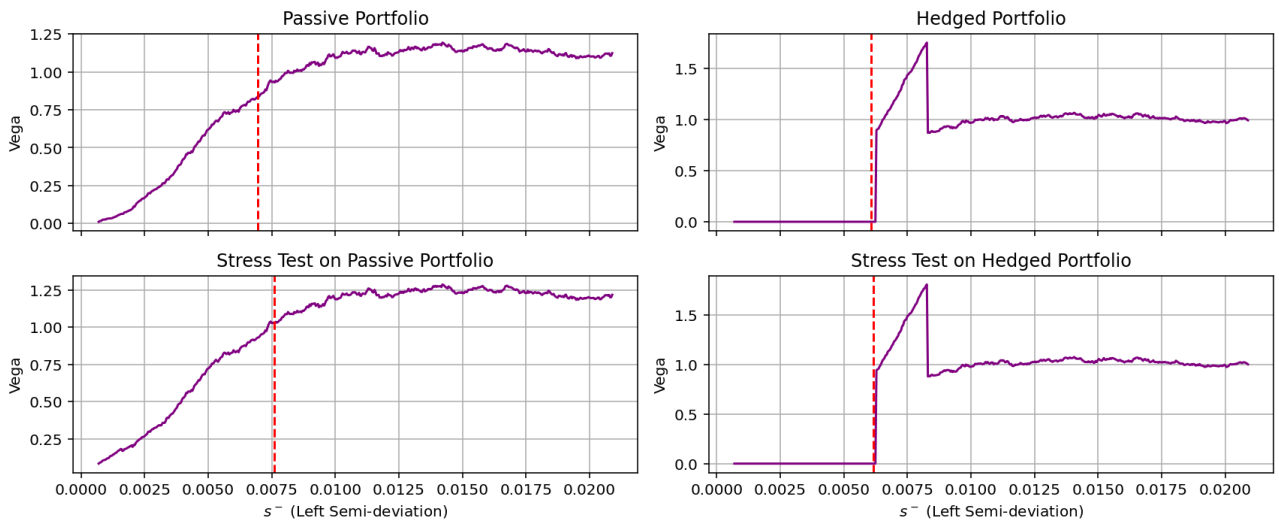


Figure 20. Graphs of V for $K = -0.032$

From the graph, it emerges that, in the passive portfolio, the stress test caused an increase in the left semi-deviation which determined an increment of $\xi(s^-, K)$ and, consequently, of the Vega fragility. On the contrary, in the hedged portfolio, the increase in the left semi-deviation was minimal and did not cause an increment in the Vega. The case reported in the graph is the one in which $K = -0.032$. For $K = -0.02$, an increase in the Vega is observed, but the differences between the Vegas with and without the stress test demonstrate that the increment of the Vega is smaller in the case of the hedged portfolio. Another observation that emerges from the graphs is the reaching of a plateau of the value of $\xi(s^-, K)$ as s^- increases. Therefore, for very high s^- , the Vega value stabilizes around 1. This implies that beyond a certain threshold of s^- and for a determined value of K , $\xi(s^-, K)$ grows linearly, that is, at a constant speed. Further increments of s^- will produce a proportional increment of $\xi(s^-, K)$. This is confirmed by the fact that the second derivative of Vega stabilizes at 0 for very large values of s^- .

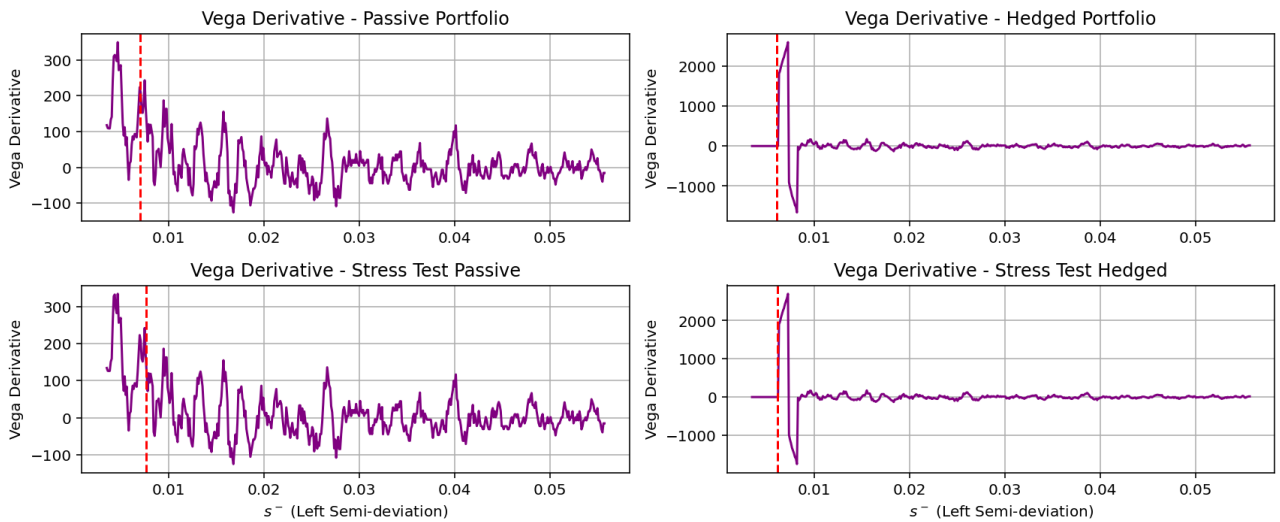


Figure 21. Second-order fragility for $K = -0.032$

The second derivative of Vega with respect to s^- represents the second-order fragility. It measures the speed with which Vega varies as s^- varies. It is observed that, for very small values of s^- , the second-order fragility marks high values.

The last aspect to analyse, in relation to fragility, is the fragility drift. As illustrated in chapter two, the fragility drift measures the speed with which the fragility separates from its original value equal to 1, when K moves away from Ω . The following figure illustrates the fragility drift of the four cases:

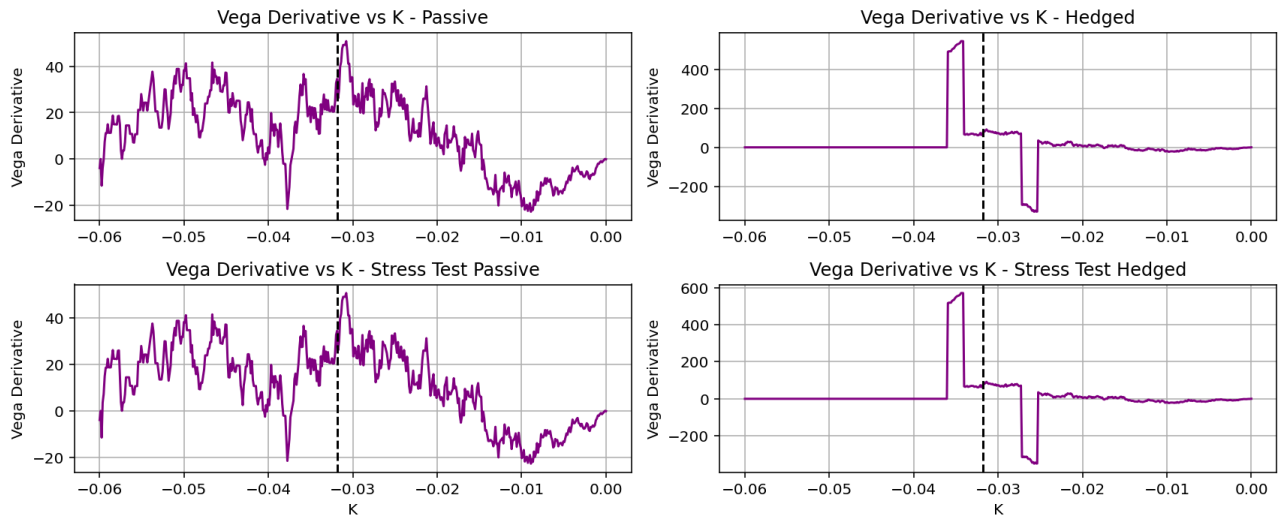


Figure 22. Fragility drift for $K = -0.032$

The graphs before and after the stress test are substantially unchanged. It is observed, furthermore, that the graph of the passive portfolio seems to react in a different manner to different levels of K , with a peak around $K \approx -0.031$. While the hedged portfolio shows high values before and after the activation level of the Put.

On the basis of these data, we can therefore affirm that introducing a Put option into the portfolio contributes to making the portfolio less sensitive to extreme negative variations of returns. The level of our protection will vary based on how much we wish to remain exposed to such risks. If, for example, we wished to cover ourselves only from the most extreme risks, we could purchase extremely OTM Put options and measure the fragility on a level $K = -0.032$ (quantile (0.05) of the S&P 500 returns). Suppose we wish to protect ourselves from returns lower than -0.05 . In this case, we obtain the following values:

Table 9: Tail metrics

Portfolios	s^-	λ	$\xi(s^-, K)$	$V(X, f_\lambda, K, s^-)$
Passive Portfolio	0.006964	1.000000	0.002490	0.836177
Hedged Portfolio	0.006719	1.000000	0.002167	0.833180
Stress test on passive portfolio	0.007614	1.000000	0.003063	0.833960
Stress test on hedged portfolio	0.006856	1.000000	0.002227	0.810057

The results show that the fragility of the hedged portfolio would be only slightly lower than that of the passive portfolio. In practice, the hedge protects only against extremely negative returns. Furthermore, the portfolio results as less robust than those analysed previously.

Let us now pass to the analysis of the right side of the distribution of returns. The analysis is substantially specular to that just performed for the left tail; for this reason, we will limit ourselves to presenting the main results. In the following tables, the values of: s^+ , $\xi(s^+, L)$, $W(X, f_\lambda, K, s^+, \Delta s)$ have been calculated. The main difference concerns the calculation of s^+ : while for s^- the CDF was utilized, for s^+ the survival function was utilized, since the integral of the right side of the CDF does not represent the equivalent of the left semi-deviation. From a graphical point of view, the difference is the following:

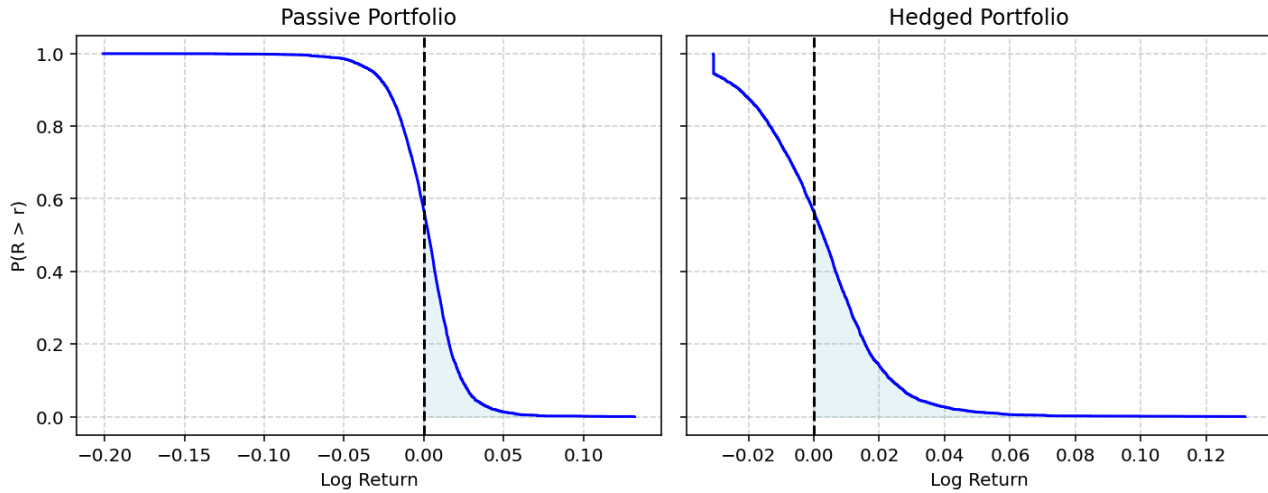


Figure 23. Survival function of the portfolios

Regarding the results, three tests were conducted for different levels of L - respectively 0.02, 0.03, 0.04 - to study "right fragility." We recall that antifragility is not the exact opposite of fragility, but an antifragile phenomenon must necessarily be robust; that is, it must have a V contained within a level deemed acceptable and a W value as high as possible. The results obtained from the analysis of W are the following:

Table 10: for $L = 0.02$ and $\Delta s = 1e - 3$

Portfolios	s^+	λ	$\xi(s^+, L)$	$W(X, f_\lambda, L, s^+)$.
Passive Portfolio	0.008457	1.000000	0.000059	0.020308
Hedged Portfolio	0.008345	0.986796	0.000056	0.019549

Table 11: For $L = 0.03$ ($\approx$ quantile $0.95 = 0.031$) and $\Delta s = 1e - 3$

Portfolios	s^+	λ	$\xi(s^+, L)$	$W(X, f_\lambda, L, s^+)$.
Passive Portfolio	0.008457	1.000000	0.000036	0.014347
Hedged Portfolio	0.008345	0.986796	0.000034	0.013719

Table 12: For $L = 0.04$ and $\Delta s = 1e - 3$

Portfolios	s^+	λ	$\xi(s^+, L)$	$W(X, f_\lambda, L, s^+)$.
Passive Portfolio	0.008457	1.000000	0.000022	0.009982
Hedged Portfolio	0.008345	0.986796	0.000021	0.009469

The tables show values of $\xi(s^+, L)$ and W that are very low when compared with those of $\xi(s^-, K)$ and V . This leads us to say that the distribution is much less exposed to the opportunities deriving from positive volatility than to the dangers of negative volatility. The expected benefit from extreme positive returns is much lower than the expected damage from extreme events. The anomaly lies in the value s^+ , which is significantly higher compared to its counterpart s^- . This means that, despite more positive returns being recorded, these are of contained magnitude, that is, to a large extent lower than the threshold L . In the case of s^- , instead, the opposite is observed: many more returns cross the threshold K . The thresholds K and L do not necessarily have to be chosen in a symmetric way; even adopting a more generous threshold L , such as $L = 0.02$, the result does not vary substantially. From the comparison between the two types of portfolios, another important datum emerges: namely, the cost of the Put entails a reduction of W . This is observed in all the analysed scenarios. The Put, when it does not enter into action, is worth zero and in fact remains only a cost. This reduces the return and causes a decrease in the sensitivity of the right tail W . This decrease is proportional to the cost of the Put, whereby it will be greater in the case in which the protection is high and smaller in the opposite case. From the following figure, one can observe the effect of the introduction of the Put on the right semi-deviation and, consequently, on the W value.

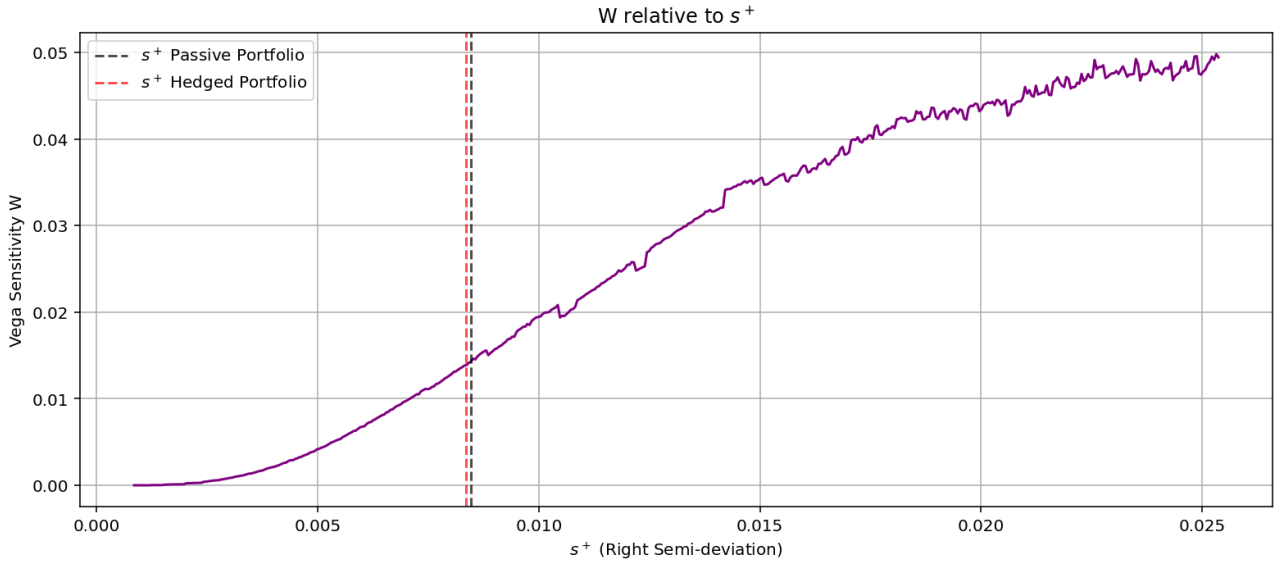


Figure 24. Right-side fragility W with respect to s^+

Analysis of Barbell Portfolios

In this paragraph, we dedicate ourselves to the analysis of Barbell portfolios. The Barbell portfolios considered include risky assets such as cryptocurrencies, small-cap stocks, and blue-chips of the American technology sector. The risk-free asset utilized is the American T-Bill. The considered period is 5 years and the examined returns are weekly. Before passing to the creation of the portfolios, for each asset the following were measured: s^- , $\xi(s^-, K)$, $V(X, f_\lambda, K, s^-, \Delta s)$ and s^+ , $\xi(s^+, L)$, $W(X, f_\lambda, L, s^+, \Delta s)$ for various values of K and L , with $\Delta s = 1e - 3$ for V and $\Delta s = 1e - 2$ for W . This difference is necessary in order to obtain more precise estimates for V and W . The results are the following:

Table 13: For $K = -0.03$ and $L = 0.03$

Asset	s^-	s^+	$\xi(s^-, K)$	$\xi(s^+, L)$	$V(X, f_\lambda, K, s^-)$	$W(X, f_\lambda, L, s^+)$
AAPL	0.01283	0.017608	0.009623	0.000279	1.324309	0.039226
MSFT	0.011905	0.015861	0.008825	0.000193	1.032445	0.029233
GOOGL	0.013520	0.017374	0.009765	0.000270	1.005502	0.032579
AMZN	0.016389	0.019594	0.012263	0.000417	1.321840	0.046900
BTC-USD	0.028904	0.038279	0.026550	0.002125	0.918538	0.105730
ETH-USD	0.037090	0.049186	0.035326	0.003571	1.009623	0.131049
LTC-USD	0.043314	0.046382	0.041504	0.003131	1.016987	0.131141
XRP-USD	0.043062	0.051811	0.040937	0.006494	1.067907	0.226169
DOGE-USD	0.046976	0.065690	0.045284	0.014112	0.963985	0.227830

NIO	0.042682	0.043823	0.041006	0.003435	1.018338	0.146670
TRMD	0.023533	0.028442	0.020962	0.001313	1.065541	0.079450
FUBO	0.060492	0.052426	0.059170	0.006052	1.035454	0.195422
AAOI	0.045613	0.049794	0.043750	0.004982	1.131458	0.175009
SPCE	0.059951	0.045896	0.058644	0.003626	1.035596	0.156042
ZM	0.028537	0.029024	0.026245	0.001393	1.093778	0.089254
RF	0.000000	0.000469	0.000000	0.000000	0.500012	0.000000

Table 14: For $K = -0.04$ and $L = 0.04$

Asset	s^-	s^+	$\xi(s^-, K)$	$\xi(s^+, L)$	$V(X, f_\lambda, K, s^-)$	$W(X, f_\lambda, L, s^+)$
AAPL	0.01283	0.017608	0.007227	0.000214	1.186875	0.033598
MSFT	0.011905	0.015861	0.007044	0.000138	1.294629	0.024348
GOOGL	0.013520	0.017374	0.008435	0.000202	0.859866	0.027834
AMZN	0.016389	0.019594	0.010039	0.000344	1.001116	0.043191
BTC-USD	0.028904	0.038279	0.025351	0.001982	1.029229	0.105219
ETH-USD	0.037090	0.049186	0.034569	0.003430	1.008449	0.130863
LTC-USD	0.043314	0.046382	0.040835	0.003008	1.096764	0.129684
XRP-USD	0.043062	0.051811	0.040005	0.006383	1.005709	0.224486
DOGE-USD	0.046976	0.065690	0.042108	0.013985	1.360231	0.227946
NIO	0.042682	0.043823	0.039820	0.003310	1.084785	0.145796
TRMD	0.023533	0.028442	0.018934	0.001212	1.189190	0.077057
FUBO	0.060492	0.052426	0.057805	0.005960	0.955585	0.193171
AAOI	0.045613	0.049794	0.042433	0.004837	1.161976	0.172957
SPCE	0.059951	0.045896	0.057429	0.003549	1.035630	0.155458
ZM	0.028537	0.029024	0.024478	0.001311	1.090537	0.084066
RF	0.000000	0.000469	0.000000	0.000000	0.500012	0.000000

Table 15: For $K = -0.05$ and $L = 0.05$

Asset	s^-	s^+	$\xi(s^-, K)$	$\xi(s^+, L)$	$V(X, f_\lambda, K, s^-)$	$W(X, f_\lambda, L, s^+)$
AAPL	0.01283	0.017608	0.006185	0.000151	0.868677	0.028519
MSFT	0.011905	0.015861	0.004779	0.000087	1.077051	0.019366
GOOGL	0.013520	0.017374	0.007023	0.000153	1.088911	0.022480
AMZN	0.016389	0.019594	0.008853	0.000259	0.635386	0.036494
BTC-USD	0.028904	0.038279	0.022727	0.001842	1.077619	0.103447

ETH-USD	0.037090	0.049186	0.032701	0.003267	1.074245	0.131798
LTC-USD	0.043314	0.046382	0.039778	0.002849	1.206318	0.127323
XRP-USD	0.043062	0.051811	0.038425	0.006257	1.274122	0.223112
DOGE-USD	0.046976	0.065690	0.040371	0.013891	0.956486	0.222369
NIO	0.042682	0.043823	0.037917	0.003176	1.178249	0.141759
TRMD	0.023533	0.028442	0.017027	0.001095	1.107191	0.073550
FUBO	0.060492	0.052426	0.057088	0.005851	0.943735	0.192793
AAOI	0.045613	0.049794	0.040913	0.004700	0.991838	0.172437
SPCE	0.059951	0.045896	0.055599	0.003408	1.700149	0.154096
ZM	0.028537	0.029024	0.023098	0.001214	1.380573	0.082303
RF	0.000000	0.000469	0.000000	0.000000	0.500012	0.000000

The values of s^- and s^+ immediately show a first distinction between asset classes: tech stocks, such as AAPL, MSFT, GOOGL, and AMZN, record thresholds s^- and s^+ on the order of a few hundred basis points ($\approx 0.011 - 0.02$), while cryptocurrencies and some small-cap securities (BTC, ETH, LTC, XRP, DOGE, FUBO, AAOI, SPCE) present much higher thresholds, with s^- reaching up to 0.06 for SPCE and FUBO and s^+ exceeding 0.05 for DOGE and XRP. This indicates that, in the considered time horizon and for weekly returns, the tails of the distributions are much wider for cryptocurrencies and for many small-caps. In practice, these assets are subject to greater uncertainty both in a positive and in a negative sense, justifying their use within Barbell portfolios.

The functions $\xi(s^-, K)$ and $\xi(s^+, L)$ bring to light a consistent behaviour as the thresholds K and L vary: as increases (passing from -0.03 to -0.05), the values of ξ calculated in correspondence with s^- generally tend to reduce. Taking AAPL as an example, $\xi(s^-, K)$ passes from 0.009623 ($K = -0.03$) to 0.007227 ($K = -0.04$) and then to 0.006185 ($K = -0.05$). An analogous pattern is observed for the majority of the securities: this suggests that the estimated sensitivity to “less extreme” losses is higher than to more extreme losses; that is, when the threshold is pushed toward the worst case, the incremental response diminishes.

The Vegas $V(X, f_\lambda, K, s^-)$ and $W(X, f_\lambda, L, s^+)$ highlight very different roles in the negative and positive portion of the distribution. V assumes in many cases values close to or higher than 1 for stocks and cryptocurrencies (for example, AAPL in the three scenarios passes from 1.324 to 1.186 to 0.868), signalling a substantial exposure “to volatility” on the negative side. For some assets, V decreases as the threshold worsens (AAPL). For others, V shows non-monotonic variations;

therefore, the effect on the negative component is sensitive to the definition of the scenario and to the choice of Δs . W , the measure of sensitivity of the positive tail, is instead marked for some assets: DOGE and XRP maintain very high W (between 0.22 and 0.23) in all three scenarios, BTC and ETH record intermediate-level W (BTC ≈ 0.10 , ETH ≈ 0.13), while blue-chips show contained W (AAPL ≈ 0.028 – 0.039 , MSFT ≈ 0.019 – 0.029). This pattern indicates that the potential for gain in the positive tails is very concentrated on cryptocurrencies and on some small-caps. This is consistent with the Barbell logic that assigns the majority of the expected positive payoff to the risky component.

The comparison between scenarios also shows a scale effect on the same metrics: by increasing the positive threshold L from 0.03 to 0.05, the values of (s^+, L) remain very small (order 10^{-4} - 10^{-3}) for all assets, signaling that sensitivity in the right tail is limited. Parallely, W tends to decrease slightly for some assets as L grows, as if the gain effect per unit of threshold were attenuated by moving the threshold further away. For example, for AAPL, W falls from 0.039226 ($L=0.03$) to 0.033598 ($L=0.04$) to 0.028519 ($L=0.05$). This behaviour suggests that large extreme gains are less frequent as the threshold increases L .

Another relevant aspect is the stability of the role of the T-Bill (RF): as expected, RF shows null $s^- = 0$, $s^+ = 0.000469$, $\xi(s^-, K)$ and $\xi(s^+, L)$, and practically null or fixed values of V and W ($V \approx 0.5$ due to finite differences), confirming the risk-absorption role that the risk-free component exerts in the barbell portfolio. In practical terms, the presence of RF reduces overall variance and mitigates the severity of the aggregate negative tails, while not offering significant contributions to the positive tails.

From the cross-sectional comparison, it furthermore emerges that some small-caps show particularly relevant asymmetric behaviours: AAOI and FUBO, for example, have high V (close to or greater than 1) and significant W (AAOI ≈ 0.17 , FUBO ≈ 0.195), which indicates that these securities contribute both to fragility in the negative tails and to the return potential in the positive tails. This dual role makes them typical candidates for the “risky tip” of the barbell if the objective is to capture rare positive events while accepting loss amplifications.

Finally, from a methodological point of view, it is appropriate to recall that V and W were estimated with different granularities of Δs ($1e-3$ for V and $1e-2$ for W): this choice improves precision in the estimation of V (more sensitive to the local shape of the left tail) while making W more robust with

respect to local variability on the positive side. The choice of a larger Δs for W is due to the scarcity of data in the right tail compared to the left tail. This implies that the differences observed between V and W are not only economic but also in part numerical: direct comparisons between the two measures must take into account the scale of differentiation. It is therefore advisable, for any operational decisions, to perform sensitivity tests on Δs and check the robustness of the resulting weights with respect to small changes in discretization.

In synthesis, the data show a clear separation between the conservative component of the portfolio (RF and blue-chips, with low s^+ and modest W) and the risky tip (cryptocurrencies and some small-caps, with higher s^+ , more marked $\xi(s^+, L)$, and high W). This concentration of the expected positive value on the few most volatile entries is precisely what justifies, at an empirical level, the choice of a Barbell. The protection of capital is maintained through RF and assets with lighter tails, and a strong but concentrated exposure is accepted toward the assets that fuel the right tail.

Let us now pass to the portfolios constructed utilizing the assets examined above:

Table 16: Types of portfolios examined

	Risk Free	Risky Portfolio
Type 1	85%	15%
Type 2	90%	10%
Type 3	95%	5%

For each type of portfolio, various options were examined to establish the weight of the risky assets. Different types of weight allocation were considered, including: Equal Weight, High Risk Equal Weight, Minimum Variance, and three allocations with different target risk-return ratios respectively (Sharpe ratio = 0.05, 0.07, 0.09). The result of the allocations is the following:

Table 17: Composition of Barbell portfolios (85% Risk-Free)

Asset	Equal Weight	High Risk	MinVar	SR ₁	SR ₂	SR ₃
AAPL	0.0100	0.0000	0.0352	0.0591	0.0648	0.0265
MSFT	0.0100	0.0000	0.0539	0.0323	0.0015	0.0015
GOOGL	0.0100	0.0000	0.0187	0.0060	0.0015	0.0015
AMZN	0.0100	0.0000	0.0015	0.0015	0.0015	0.0015
BTC-USD	0.0100	0.0000	0.0015	0.0037	0.0147	0.0053

ETH-USD	0.0100	0.0000	0.0015	0.0015	0.0059	0.0379
LTC-USD	0.0100	0.0000	0.0015	0.0015	0.0015	0.0015
XRP-USD	0.0100	0.0300	0.0015	0.0015	0.0015	0.0015
DOGE-USD	0.0100	0.0300	0.0015	0.0067	0.0201	0.0388
NIO	0.0100	0.0000	0.0015	0.0015	0.0015	0.0015
TRMD	0.0100	0.0000	0.0257	0.0288	0.0295	0.0266
FUBO	0.0100	0.0300	0.0015	0.0015	0.0015	0.0015
AAOI	0.0100	0.0300	0.0015	0.0015	0.0015	0.0015
SPCE	0.0100	0.0300	0.0015	0.0015	0.0015	0.0015
ZM	0.0100	0.0000	0.0015	0.0015	0.0015	0.0015
RF	0.8500	0.8500	0.8500	0.8500	0.8500	0.8500

Table 18: Composition of Barbell portfolios (90% Risk-Free)

Asset	Equal Weight	High Risk	MinVar	SR ₁	SR ₂	SR ₃
AAPL	0.0067	0.0000	0.0235	0.0394	0.0432	0.0177
MSFT	0.0067	0.0000	0.0359	0.0215	0.0010	0.0010
GOOGL	0.0067	0.0000	0.0125	0.0040	0.0010	0.0010
AMZN	0.0067	0.0000	0.0010	0.0010	0.0010	0.0010
BTC-USD	0.0067	0.0000	0.0010	0.0024	0.0098	0.0035
ETH-USD	0.0067	0.0000	0.0010	0.0010	0.0039	0.0252
LTC-USD	0.0067	0.0000	0.0010	0.0010	0.0010	0.0010
XRP-USD	0.0067	0.0200	0.0010	0.0010	0.0010	0.0010
DOGE-USD	0.0067	0.0200	0.0010	0.0045	0.0134	0.0259
NIO	0.0067	0.0000	0.0010	0.0010	0.0010	0.0010
TRMD	0.0067	0.0000	0.0171	0.0192	0.0197	0.0177
FUBO	0.0067	0.0200	0.0010	0.0010	0.0010	0.0010
AAOI	0.0067	0.0200	0.0010	0.0010	0.0010	0.0010
SPCE	0.0067	0.0200	0.0010	0.0010	0.0010	0.0010
ZM	0.0067	0.0000	0.0010	0.0010	0.0010	0.0010
RF	0.9000	0.9000	0.9000	0.9000	0.9000	0.9000

Table 19: Composition of Barbell portfolios (95% Risk-Free)

Asset	Equal Weight	High Risk	MinVar	SR ₁	SR ₂	SR ₃
AAPL	0.0033	0.0000	0.0117	0.0197	0.0216	0.0088

MSFT	0.0033	0.0000	0.0180	0.0108	0.0005	0.0005
GOOGL	0.0033	0.0000	0.0062	0.0020	0.0005	0.0005
AMZN	0.0033	0.0000	0.0005	0.0005	0.0005	0.0005
BTC-USD	0.0033	0.0000	0.0005	0.0012	0.0049	0.0018
ETH-USD	0.0033	0.0000	0.0005	0.0005	0.0020	0.0126
LTC-USD	0.0033	0.0000	0.0005	0.0005	0.0005	0.0005
XRP-USD	0.0033	0.0100	0.0005	0.0005	0.0005	0.0005
DOGE-USD	0.0033	0.0100	0.0005	0.0022	0.0067	0.0129
NIO	0.0033	0.0000	0.0005	0.0005	0.0005	0.0005
TRMD	0.0033	0.0000	0.0086	0.0096	0.0098	0.0089
FUBO	0.0033	0.0100	0.0005	0.0005	0.0005	0.0005
AAOI	0.0033	0.0100	0.0005	0.0005	0.0005	0.0005
SPCE	0.0033	0.0100	0.0005	0.0005	0.0005	0.0005
ZM	0.0033	0.0000	0.0005	0.0005	0.0005	0.0005
RF	0.9500	0.9500	0.9500	0.9500	0.9500	0.9500

From the analysis of the weights, it emerges that the allocations based on the Markowitz method tend to concentrate the risk excessively on a few securities. This behaviour, although consistent with the objective of minimizing the overall variance of the risky portfolio, reduces the effectiveness of the Barbell strategy as a whole. Moreover, the chosen securities are the same ones that in the previous analysis show the lowest values of W and values of V not substantially lower than those of the riskier assets.

The logic of the Barbell requires that the risky component be widely diversified within a group of high-variance bets, with the objective of capturing extreme positive events, without compromising the overall stability guaranteed by the risk-free share.

The Markowitzian optimization approach, on the contrary, penalizes the tails and tends to underweight or exclude assets with high variance, thus nullifying the function of the risky component. It follows that the risk of the risky portfolio results as more concentrated and less antifragile. The performance becomes dependent on a few assets, and the decline of just one of them could generate a loss superior to that which one would have from a simultaneous decline of multiple diversified assets.

In synthesis, Markowitz optimization results as inefficient in the Barbell context, where the objective is not to minimize variance, but to maximize asymmetric exposure to positive tail opportunities. Let us now analyse the distributions of the Equal Weight portfolios to analyse the effect produced by the increase in the weight of the Risk-Free asset within the portfolios:

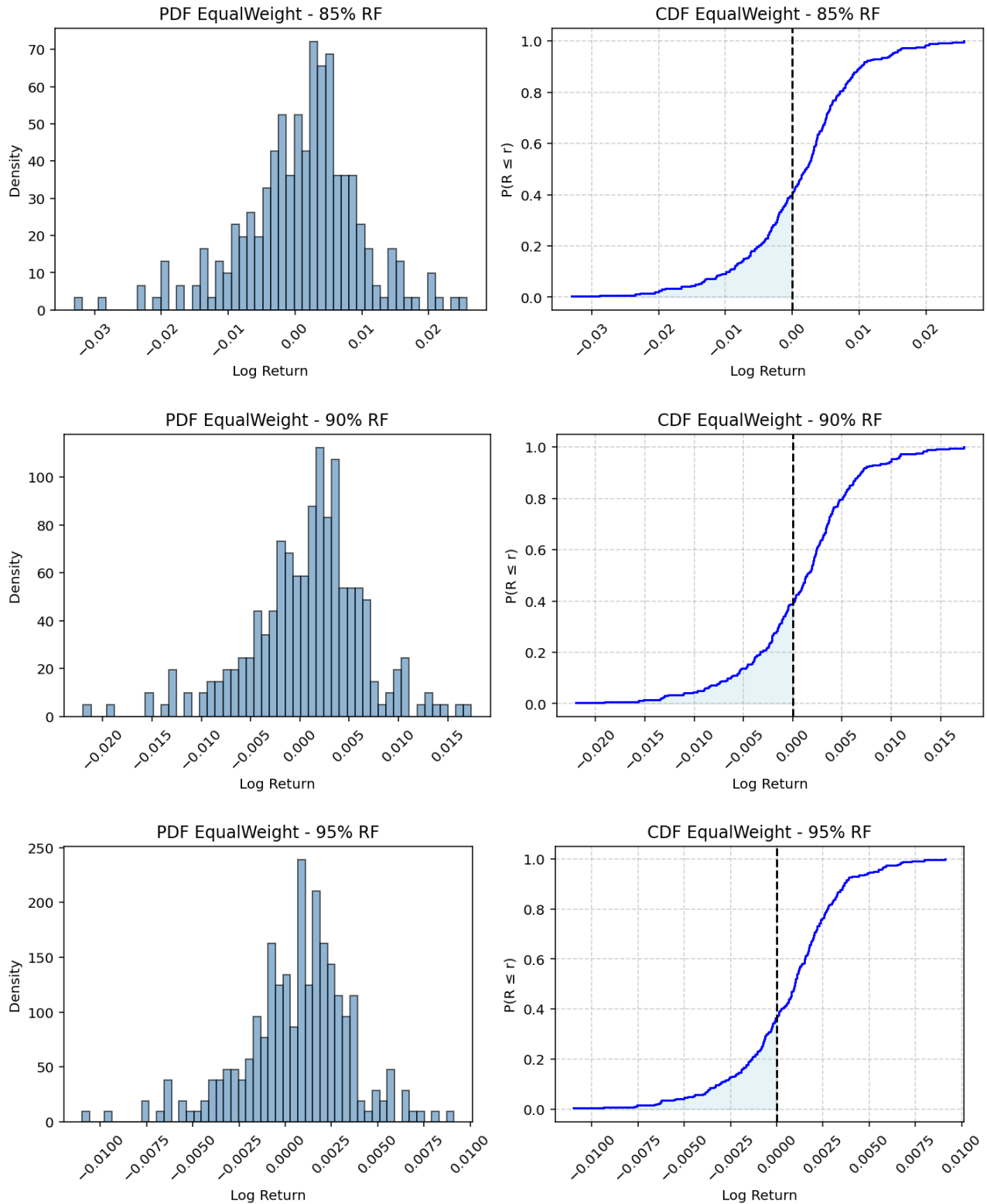


Figure 25. PDF and CDF of Barbell portfolios

What we observe in these graphs is a decrease in the fragility of the distribution due to the addition of the risk-free asset. By increasing the share of T-bills, the tails of the portfolio return distribution shorten. These figures are in line with the robust distribution shown in the previous chapter.

Let us now analyse the results obtained by calculating V and W for the various types of portfolios and for various levels of K and L :

Portfolios of the first type 85% RF:

Table 20: For $K = -0.005$ and $L = 0.005$ with $\Delta s = 1e - 3$

Portfolio	s^-	s^+	$\xi(s^-, K)$	$\xi(s^+, L)$	$V(X, f_\lambda, K, s^-)$	$W(X, f_\lambda, L, s^+)$
EqualWeight	0.002844	0.003807	0.002357	0.000014	1.045174	0.008697
HighRisk	0.004960	0.005644	0.004525	0.000050	1.062332	0.017172
MinVar	0.001380	0.002416	0.000869	0.000003	0.971025	0.003860
SR_1	0.001405	0.002554	0.000952	0.000004	1.018735	0.004318
SR_2	0.001939	0.003387	0.001468	0.000011	1.118187	0.007143
SR_3	0.003279	0.005177	0.002776	0.000043	1.056787	0.015950

Table 21: For $K = -0.01$ and $L = 0.01$ with $\Delta s = 1e - 3$

Portfolio	s^-	s^+	$\xi(s^-, K)$	$\xi(s^+, L)$	$V(X, f_\lambda, K, s^-)$	$W(X, f_\lambda, L, s^+)$
EqualWeight	0.002844	0.003807	0.001562	0.000008	1.050951	0.006236
HighRisk	0.004960	0.005644	0.003693	0.000040	1.163073	0.015946
MinVar	0.001380	0.002416	0.000327	0.000000	0.626047	0.001111
SR_1	0.001405	0.002554	0.000430	0.000001	0.657404	0.001200
SR_2	0.001939	0.003387	0.000797	0.000006	0.874259	0.004605
SR_3	0.003279	0.005177	0.002048	0.000035	1.035520	0.014673

Table 22: For $K = -0.02$ and $L = 0.02$ with $\Delta s = 1e - 3$

Portfolio	s^-	s^+	$\xi(s^-, K)$	$\xi(s^+, L)$	$V(X, f_\lambda, K, s^-)$	$W(X, f_\lambda, L, s^+)$
EqualWeight	0.002844	0.003807	0.000495	0.000001	0.623488	0.001297
HighRisk	0.004960	0.005644	0.001591	0.000024	0.977449	0.011051
MinVar	0.001380	0.002416	0.000000	0.000000	0.248113	0.000000
SR_1	0.001405	0.002554	0.000000	0.000000	0.264715	0.000000
SR_2	0.001939	0.003387	0.000280	0.000001	0.469386	0.000605

SR_3	0.003279	0.005177	0.000824	0.000020	0.700004	0.008918
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Table 23: For $K = -0.02$ and $L = 0.02$ with $\Delta s = 1e - 3$

Portfolio	s^-	s^+	$\xi(s^-, K)$	$\xi(s^+, L)$	$V(X, f_\lambda, K, s^-)$	$W(X, f_\lambda, L, s^+)$
EqualWeight	0.002844	0.003807	0.000127	0.000000	0.280907	0.000000
HighRisk	0.004960	0.005644	0.000688	0.000012	0.417392	0.006838
MinVar	0.001380	0.002416	0.000000	0.000000	0.000000	0.000000
SR_1	0.001405	0.002554	0.000000	0.000000	0.120313	0.000000
SR_2	0.001939	0.003387	0.000000	0.000000	0.212031	0.000000
SR_3	0.003279	0.005177	0.000446	0.000011	0.484276	0.007109

Table 24: For $K = -0.04$ and $L = 0.04$ with $\Delta s = 1e - 3$

Portfolio	s^-	s^+	$\xi(s^-, K)$	$\xi(s^+, L)$	$V(X, f_\lambda, K, s^-)$	$W(X, f_\lambda, L, s^+)$
EqualWeight	0.002844	0.003807	0.000000	0.000000	0.085734	0.000000
HighRisk	0.004960	0.005644	0.000159	0.000003	0.339269	0.003183
MinVar	0.001380	0.002416	0.000000	0.000000	0.000000	0.000000
SR_1	0.001405	0.002554	0.000000	0.000000	0.000000	0.000000
SR_2	0.001939	0.003387	0.000000	0.000000	0.077117	0.000000
SR_3	0.003279	0.005177	0.000161	0.000004	0.290986	0.004225

For values of K and L exceeding 0.05 in absolute terms, the values of V and W are zero for all portfolios.

Portfolios of the first type 90% RF:

Table 25: For $K = -0.005$ and $L = 0.005$ with $\Delta s = 1e - 3$

Portfolio	s^-	s^+	$\xi(s^-, K)$	$\xi(s^+, L)$	$V(X, f_\lambda, K, s^-)$	$W(X, f_\lambda, L, s^+)$
EqualWeight	0.001838	0.002636	0.001299	0.000005	1.063100	0.004838
HighRisk	0.003240	0.003852	0.002783	0.000021	1.141078	0.010404
MinVar	0.000871	0.001718	0.000308	0.000001	0.718694	0.001630
SR_1	0.000890	0.001812	0.000405	0.000001	0.803226	0.001792
SR_2	0.001240	0.002361	0.000728	0.000004	0.911242	0.004206
SR_3	0.002121	0.003543	0.001550	0.000018	1.074685	0.010143

Table 26: For $K = -0.01$ and $L = 0.01$ with $\Delta s = 1e - 3$

Portfolio	s^-	s^+	$\xi(s^-, K)$	$\xi(s^+, L)$	$V(X, f_\lambda, K, s^-)$	$W(X, f_\lambda, L, s^+)$
EqualWeight	0.001838	0.002636	0.000652	0.000002	0.816335	0.002333
HighRisk	0.003240	0.003852	0.001712	0.000014	1.142891	0.008257
MinVar	0.000871	0.001718	0.000124	0.000000	0.391105	0.000000
SR_1	0.000890	0.001812	0.000138	0.000000	0.449586	0.000000
SR_2	0.001240	0.002361	0.000226	0.000001	0.602023	0.001618
SR_3	0.002121	0.003543	0.000907	0.000012	0.880246	0.008266

Table 27: For $K = -0.02$ and $L = 0.02$ with $\Delta s = 1e - 3$

Portfolio	s^-	s^+	$\xi(s^-, K)$	$\xi(s^+, L)$	$V(X, f_\lambda, K, s^-)$	$W(X, f_\lambda, L, s^+)$
EqualWeight	0.001838	0.002636	0.000085	0.000000	0.370310	0.000000
HighRisk	0.003240	0.003852	0.000457	0.000005	0.511684	0.004343
MinVar	0.000871	0.001718	0.000000	0.000000	0.132807	0.000000
SR_1	0.000890	0.001812	0.000000	0.000000	0.146334	0.000000
SR_2	0.001240	0.002361	0.000000	0.000000	0.168041	0.000000
SR_3	0.002121	0.003543	0.000297	0.000005	0.402007	0.004349

For values of K and L exceeding 0.03 in absolute terms, the values of V and W are zero for all portfolios.

Portfolios of the first type 95% RF:

Table 28: For $K = -0.005$ and $L = 0.005$ with $\Delta s = 1e - 3$

Portfolio	s^-	s^+	$\xi(s^-, K)$	$\xi(s^+, L)$	$V(X, f_\lambda, K, s^-)$	$W(X, f_\lambda, L, s^+)$
EqualWeight	0.000839	0.001472	0.000321	0.000000	0.680293	0.001072
HighRisk	0.001524	0.002064	0.000800	0.000004	0.972849	0.004120
MinVar	0.000378	0.001036	0.000061	0.000000	0.475973	0.000000
SR_1	0.000395	0.001090	0.000068	0.000000	0.522228	0.000000
SR_2	0.000559	0.001354	0.000112	0.000000	0.600510	0.000748
SR_3	0.000973	0.001918	0.000407	0.000003	0.736359	0.003228

Table 29: For $K = -0.01$ and $L = 0.01$ with $\Delta s = 1e - 3$

Portfolio	s^-	s^+	$\xi(s^-, K)$	$\xi(s^+, L)$	$V(X, f_\lambda, K, s^-)$	$W(X, f_\lambda, L, s^+)$
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EqualWeight	0.000839	0.001472	0.000042	0.000000	0.372565	0.000000
HighRisk	0.001524	0.002064	0.000227	0.000001	0.449366	0.002012
MinVar	0.000378	0.001036	0.000000	0.000000	0.236172	0.000000
SR_1	0.000395	0.001090	0.000000	0.000000	0.225309	0.000000
SR_2	0.000559	0.001354	0.000000	0.000000	0.322156	0.000000
SR_3	0.000973	0.001918	0.000148	0.000001	0.432430	0.001336

For values of K and L exceeding 0.02 in absolute terms, the values of V and W are zero for all portfolios.

Analysis of results:

The progressive increase of the risk-free share from 85% to 95% significantly reduces the width of the tails (s^- and s^+) and the intensity of the contributions ξ associated with extreme loss and gain levels. Intuitively, the volatility of the tails compresses with the increase of the T-bill share, and the distribution of overall returns becomes increasingly narrow, reducing both the vulnerability to negative shocks and the possibility of extreme benefits.

In portfolios with 85% risk-free, the tails result as relatively wide ($s^- \approx 0.0028 - 0.0049$ and $s^+ \approx 0.0038 - 0.0056$). The values of $\xi(s^-, K)$ and $\xi(s^+, L)$ indicate the presence of returns that exceed the thresholds K and L . If we look at the different portfolios, we note in general that the MinVar (Minimum Variance) portfolios show the lowest values of V and W . This is consistent with the idea of constructing less risky portfolios, but in the case of Barbell portfolios, the objective is not to minimize exposure to risks, but on the contrary, to maximize it. The risk control of the strategy is based on the share of capital invested in the risk-free security. For this reason, the Equal Weight and High Risk portfolios result as the most coherent with the Barbell strategy. The diversified exposure to multiple risky securities allows for preserving the possibility of extreme gains without compromising the overall stability of the portfolio.

When the risk-free share passes to 90%, an overall downsizing of s^- and s^+ is observed (approximately halved) and a proportional reduction of the components $\xi(s^-, K)$ and $\xi(s^+, L)$. This entails a compression of the distribution, consistent with a more defensive portfolio. The risk of extreme loss is reduced, but the potential for exceptional gains also decreases. The metrics V and W settle on average between 0.8 and 1.1, showing a still acceptable balance between robustness and the capacity to capture tail opportunities. However, in the MinVar, SR_1, SR_2

portfolios, the values of W remain null or negligible: this indicates a reduction in sensitivity to positive events, i.e., a loss of the antifragile dimension of the Barbell. In the High Risk and SR_3 portfolios, instead, W maintains values between 0.004 and 0.01, against a reduced V compared to the 85% Risk-Free portfolios.

Moving to 95% risk-free, the behavior becomes clearly dominated by the safe component: s^- and s^+ fall below 0.002. For the $|0.005|$ threshold of K and L , $\xi(s^-, K)$ and (s^+, L) become almost null in all portfolios, with the exception of the two most aggressive and least diversified (High Risk and SR_3). V remains comprised between 0.45 and 0.97, while W is reduced by half compared to the previous case. The Equal Weight preserves a relatively stable profile ($V \approx 0.68$, $W \approx 0.001$), while High Risk still shows a slightly thicker right tail ($V \approx 0.97$, $W \approx 0.004$). However, in the Markowitzian portfolios, the values of W zero out completely.

This highlights that beyond a certain threshold of prudence, the Barbell effect cancels itself out: the total protection provided by the risk-free part neutralizes even the potential for extreme gains, causing the strategy to lose its antifragile nature.

Let us now analyse the connection between what has been observed and the compositions of the portfolios. By observing the weights of the individual assets, the structural difference between the strategies clearly emerges:

- Equal Weight distributes the risk uniformly across all assets, including the most volatile ones (BTC, ETH, DOGE), guaranteeing a good exposure to the right tails and therefore a form of natural antifragility.
- High Risk concentrates the equity and crypto portion on a few assets with higher variance, increasing the sensitivity to the tails but also the overall fragility of the portfolio.
- MinVar, SR_1 , and SR_2 , instead, drastically reduce the weights of the riskiest assets, privileging stable securities such as AAPL, MSFT, or TRMD. This leads to a more robust but also flatter profile: the portfolio protects itself from negative shocks, but no longer benefits from positive ones.

The combined observation of the values of V and W with the compositions indicates that the logic of Markowitz and that of the Barbell are profoundly in conflict: the former aims to eliminate volatility, the latter to exploit it.

In consistency with what was stated by Taleb in his article, the Barbell strategy results as being robust but not antifragile. It is capable of limiting the problem of heavy tails, but not of exploiting negative variability to improve. Furthermore, the Barbell strategy hypothesizes that the risk-free instrument is truly such. If, in our case, the American state was to reveal itself as insolvent, the loss of the portfolio would be enormous.

Conclusions

In this chapter, an empirical application of the definitions of fragility, robustness, and antifragility introduced in the previous chapter has been presented, through the analysis of three different investment strategies. The obtained results confirm the theories of Taleb, showing how passive investment behaves as a fragile phenomenon, Barbell portfolios as robust, and the hedging strategy with Put options manifests antifragile characteristics, capable of limiting extreme losses and exploiting positive gains.

The numerical analysis of Vega sensitivities has highlighted the greater vulnerability of passive investment to extreme events, while hedging with options has demonstrated an effective mitigation of left fragility and an efficient exposure to the benefits of right fragility. Although the antifragile strategy results as being theoretically more effective, its operational complexity and transaction costs may limit its practical applicability. Barbell portfolios, while representing a less complex middle ground, do not allow for the full exploitation of the potential benefits deriving from positive fragility and are exposed to the risk of unforeseen events regarding the securities considered risk-free. Finally, the passive approach, traditionally recommended for the retail investor, confirms itself as the most fragile in terms of vulnerability to extreme risks, calling attention to the necessity of a rethinking of investment strategies in light of the complexity of markets and of rare but impactful events.

This empirical analysis thus lays the foundations for a broader reflection on risk management and on the design of resilient portfolios, opening the way to themes that will be developed in the concluding considerations.

Conclusions

The declared objective of this thesis was to verify the applicability of the mathematical definitions proposed by Taleb to markets and investment portfolios. To this end, for introductory purposes, the main discoveries, theories, and methods of market analysis were presented within Chapter 1 and, in opposition to these, almost as an antithesis, the criticisms and the principal reflections presented in previous writings by Taleb himself.

In the following chapter, the mathematical formulations that describe and allow us to classify and distinguish between fragile, robust, or antifragile phenomena were analysed. The approach utilized in the formulation of these theories is innovative, because at the basis of this process there is no hypothesis regarding the theoretical distribution that characterizes a determined phenomenon. On the contrary, the distinction between these three types of phenomena is based on the empirical distribution of the data. The disadvantage of such a method is clearly the necessity of having many data points available in order to validate one's analysis. Recalling what was said in Chapter 1, phenomena belonging to Extremistan necessitate a great many observations before the distribution completely reveals its properties. In this context, the analysis of financial returns allows us to correctly apply such definitions to many data points of financial instruments.

Within Chapter 3, a detailed empirical application of the definitions of fragility, robustness, and antifragility, as formalized in Chapter 2, was developed. The investigation examined three main strategic typologies, permitting the verification of the response of return distributions with respect to the structural characteristics of different investment approaches in financial markets. The obtained results confirm what Taleb theorizes in his works: the empirical distribution of passive investment returns behaves as a fragile phenomenon, Barbell portfolios show robustness, while the hedging strategy with Put options highlights antifragile properties, capable of both absorbing negative shocks and exploiting positive ones.

From the numerical comparison of the Vega sensitivities as the stress threshold varies, it emerges that the passive investment is the one most vulnerable to critical events, followed by the Barbell strategy, while the hedging with options results as being the most effective in mitigating extreme fragility. The antifragile strategy indeed allows for the zeroing out of sensitivity to damages in the presence of adverse scenarios, as suggested by Taleb. From the analysis of the right-side sensitivity, it results

instead that the capacity to benefit from favourable events is maximal for the passive investment, intermediate for the hedging, and minimal for the Barbell. The only strategy capable of limiting left fragility V and of exploiting right fragility W results as being the hedging strategy, confirming its antifragility. Barbell portfolios, on the contrary, result as not being capable of exploiting the right fragility of the distribution and therefore are not ascribable as antifragile phenomena, but only as robust.

Despite such results highlighting the potential superiority of dynamic hedging, it must be remembered that this strategy presents operational complexities, significant transaction costs, and requires continuous vigilance. The analysis does not take into account management costs and transfer costs, which would lower the sensitivity W and increase the threshold K beyond which the zeroing of the Vega occurs, reducing the effectiveness of the protection. For such reasons, antifragile strategies might not be economically convenient.

The Barbell strategy results as being a less complicated option, halfway between the passive investment and the hedging strategy. To function, it requires a high number of bets. Such an approach ensures protection but does not allow for the exploitation of the full positive potential due to the limited amount invested in risky assets. The weak point of such a strategy is the eventuality that an unforeseen event hits the risk-free security. Taleb himself has often highlighted how government bonds are not truly to be considered as risk-free instruments.

Finally, the passive approach, often presented as the optimal choice for the small saver, results as being the most fragile in the face of extreme events. This is despite historical data demonstrating a certain security regarding such a strategy, suggesting that in the long period this is the best alternative. To metaphorically visualize the weakness of such an approach, I briefly report the example of the turkey on Thanksgiving Day. The turkey lives a life without worries in the farm; it is given food and water to make it grow healthily, so it thinks that it will continue this way indefinitely. But the turkey does not know when Thanksgiving Day arrives. A single event has definitively conditioned the life of the turkey. This event is a Black Swan. So far, market declines and economic crises have proved temporary and subsequently the market has returned to marking new highs. Passive investment presupposes that this process repeats itself indefinitely, but there is no guarantee of this.

From the conducted analyses, three other considerations worthy of further investigation emerge. The first concerns the capacity of such an approach to evaluate the effectiveness of a hedge with respect to tail events. The analysis of the hedged portfolio shows that the Put option is capable of zeroing out the sensitivity to extreme events. This result varies according to the protection bought and according to the established threshold K . In this regard, we could say that the protection is effective for a determined threshold K if it zeroes out the Vega sensitivity of the portfolio. The second consideration concerns instead the assets included in the Barbell portfolios. In this case, the analysis demonstrates that assets generally considered riskier, as they are subject to greater volatility, are also the assets that record higher values of V and W , proving to be the most exposed to tail risks and benefits. The final consideration concerns portfolio optimization through the Markowitz method. Such an approach induces a concentration of risk in a few assets considered less risky. In the context of the Barbell strategy, the non-risk-free assets selected are all highly risky. In this case, therefore, the concentration of risk in a single asset lowers the volatility level of the portfolio but decreases the number of bets. The portfolio return will therefore depend on the performance of a few assets. This contrasts with the principle of diversification and induces a notable increase in real riskiness, masking such a datum under a lower level of volatility.

The evaluation metrics for sensitivity to tail events proposed by Taleb are not, however, without flaws. The functions V and W necessitate levels of K and L defined subjectively. The choice of K and L , based on the empirical distribution, will render the results non-comparable transversally. The degree of subjectivity required in the definition of these thresholds considerably influences the values of V and W . A further difficulty is given by the necessity to adopt, in order to calculate the Vega, the finite difference method. Such a method adds a further element of subjectivity and difficulty in comparing results, given by the choice of the difference Δs . The high degree of subjectivity therefore makes it difficult to define universally valid sensitivity values and makes the implementation within a regulatory framework aimed at integrating such measures of tail risk into risk management equally complicated.

This work intends to propose a different perspective in the study and management of financial risks, compared to the classical methodology adopted in practice, in order to be able to face the complexity of financial markets from various points of view, increasing the tools at our disposal.

Appendix 1. Python Code

The present appendix reports the principal functions implemented in Python utilized for the data processing and the production of the outputs presented in the empirical analysis. The blocks relating to graphical visualization and the formatting of results have been omitted, maintaining exclusively the components essential for the replicability of the analysis.

The complete code, inclusive of graphical sections and auxiliary scripts, is available in the public GitHub repository:

<https://github.com/gabrielenicolasi01/Master-Thesis-.git>

In the execution of the code, pay particular attention to the thresholds K , L and to the choice of the Δs .

```
#####  
# LIBRARIES AND PACKAGES  
#####  
  
import yfinance as yf  
import numpy as np  
import pandas as pd  
import matplotlib.pyplot as plt  
import cvxpy as cp  
  
from statsmodels.distributions.empirical_distribution import ECDF  
  
import warnings  
warnings.filterwarnings('ignore', category=RuntimeWarning)  
  
#####  
# FUNCTIONS FOR  $s^-$ ,  $s^+$ ,  $\Xi(S^-,K)$ ,  $\Xi(S^+,L)$   $V$  and  $W$   
#####  
  
def s_minus(data, Omega):  
    arr = np.asarray(data)  
    return np.mean(np.maximum(Omega - arr, 0.0))  
  
def lambda_from_s_empirical(s_empirical, s_baseline, eps=0):  
    return s_empirical / (s_baseline + eps)  
  
def xi_K(data, lambda_s, Omega, K):  
    arr = np.asarray(data)  
    scaled_data = Omega + lambda_s * (arr - Omega)  
    x_sorted = np.sort(scaled_data)  
    n = len(x_sorted)  
    F_K = np.sum(x_sorted <= K) / n
```

```

mask = x_sorted <= K
x_below = x_sorted[mask]
m = len(x_below)

if m == 0:
    integral = 0.0
elif m == 1:
    integral = 0.0
else:
    widths = np.diff(x_below)
    heights = np.arange(1, m) / n
    integral = np.sum(widths * heights)
if m > 0:
    integral += (K - x_below[-1]) * (m / n)
return (Omega - K) * F_K + integral

def xi_K_on_baseline(baseline_data, lambda_s, Omega, K):
    return xi_K(baseline_data, lambda_s, Omega, K)

def vega_tail_baseline(baseline_data, s_target, s_baseline, Omega, K, Delta_s=1e-4):
    lam_plus = lambda_from_s_empirical(s_target + Delta_s, s_baseline)
    lam_minus = lambda_from_s_empirical(max(s_target - Delta_s, 0.0), s_baseline)
    xi_plus = xi_K_on_baseline(baseline_data, lam_plus, Omega, K)
    xi_minus = xi_K_on_baseline(baseline_data, lam_minus, Omega, K)
    V = (xi_plus - xi_minus) / (2 * Delta_s)
    return V

def vega_derivative_s_minus(data, s_minus_target, s_baseline, Omega, K, Delta_s):
    lam_plus = lambda_from_s_empirical(s_minus_target + Delta_s, s_baseline)
    lam_0 = lambda_from_s_empirical(s_minus_target, s_baseline)
    lam_minus = lambda_from_s_empirical(max(s_minus_target - Delta_s, 0.0), s_baseline)
    xi_plus = xi_K_on_baseline(data, lam_plus, Omega, K)
    xi_0 = xi_K_on_baseline(data, lam_0, Omega, K)
    xi_minus = xi_K_on_baseline(data, lam_minus, Omega, K)
    sec = (xi_plus - 2.0 * xi_0 + xi_minus) / (Delta_s ** 2)
    return sec

def vega_derivative_k(data, s_minus_target, Omega, K, Delta_K=1e-4):
    V_plus = vega_tail_baseline(data, s_minus_target, s_minus_target, Omega, K + Delta_K, Delta_K)
    V_minus = vega_tail_baseline(data, s_minus_target, s_minus_target, Omega, K - Delta_K, Delta_K)
    derivative = (V_plus - V_minus) / (2 * Delta_K)
    return derivative

def calculate_vega_vs_k_values(data, s_minus_target, Omega, K_vals, Delta_s=1e-4):
    vega_vals = []
    for K in K_vals:
        vega = vega_tail_baseline(data, s_minus_target, s_minus_target, Omega, K, Delta_s)
        vega_vals.append(vega)
    return np.array(vega_vals)

def integrated_vega_vs_k_discrete(data, s_minus_target, Omega, K_min, K_max, Delta_s=1e-4, n_points=500):
    K_vals = np.linspace(K_min, K_max, n_points)
    vega_vals = calculate_vega_vs_k_values(data, s_minus_target, Omega, K_vals, Delta_s)
    integral = np.trapz(vega_vals, K_vals)
    return integral

def calculate_vega_vs_s_minus(data, s_minus_vals, Omega, K, Delta_s=1e-4):
    vega_vals = []
    for s_minus in s_minus_vals:
        vega = vega_tail_baseline(data, s_minus, s_minus, Omega, K, Delta_s)
        vega_vals.append(vega)
    return np.array(vega_vals)

```

```

def integrated_vega_vs_s_minus(data, Omega, K, s_min, s_max, Delta_s=1e-4, n_points=500):
    s_minus_vals = np.linspace(s_min, s_max, n_points)
    vega_vals = calculate_vega_vs_s_minus(data, s_minus_vals, Omega, K, Delta_s)
    integral = np.trapz(vega_vals, s_minus_vals)
    return integral

def s_plus(data, Omega):
    arr = np.asarray(data)
    return np.mean(np.maximum(arr - Omega, 0.0))

def lambda_from_s_plus_empirical(s_empirical, s_baseline, eps=0):
    return s_empirical / (s_baseline + eps)

def xi_plus(data, lambda_s, Omega, L, H):
    data = np.asarray(data)
    scaled_data = Omega + lambda_s * (data - Omega)
    x_sorted = np.sort(scaled_data)
    ecdf = ECDF(x_sorted)

    mask = (x_sorted >= L) & (x_sorted <= H)
    x_in_range = x_sorted[mask]

    if len(x_in_range) == 0:
        return 0.0

    surv = 1 - ecdf(x_in_range)

    diff = np.diff(x_in_range)

    heights = surv[:-1]

    integral = np.sum(diff * (x_in_range[:-1] - Omega) * heights)

    return integral

def xi_plus_on_baseline(baseline_data, lambda_s, Omega, L, H):
    return xi_plus(baseline_data, lambda_s, Omega, L, H)

def w_vega_right_baseline(baseline_data, s_target, s_baseline, Omega, L, H, Delta_s=1e-4):
    lam_plus = lambda_from_s_plus_empirical(s_target + Delta_s, s_baseline)
    lam_minus = lambda_from_s_plus_empirical(max(s_target - Delta_s, 0.0), s_baseline)
    xi_plus_val = xi_plus_on_baseline(baseline_data, lam_plus, Omega, L, H)
    xi_minus_val = xi_plus_on_baseline(baseline_data, lam_minus, Omega, L, H)
    W = (xi_plus_val - xi_minus_val) / (2 * Delta_s)
    return W

#####
#                                     PORTFOLIO OPTIMIZATION FUNCTIONS
#####

# Portfolio optimization functions
def solve_portfolio(constraints_func, objective_func):
    w = cp.Variable(n_assets)
    prob = cp.Problem(objective_func(w), constraints_func(w))
    prob.solve(solver=cp.SCS)
    if w.value is None:
        print("Problem not solved. Weights = NaN")
        return pd.Series(np.nan, index=assets_returns.columns)
    w_val = np.array(w.value)
    w_val = w_val / w_val.sum() # normalize for safety
    return pd.Series(w_val * risk_asset_weight, index=assets_returns.columns)

```

```

# Minimum Variance
def objective_minvar(w): return cp.Minimize(cp.quad_form(w, Sigma))
def constraints_minvar(w): return [cp.sum(w) == 1, w >= min_weight_per_asset]

#####
#                                     DATA DOWNLOADING AND ANALYSIS
#####

# Download S&P 500 weekly data
ticker = yf.Ticker("^GSPC") # modify ticker if you want to change asset
data = ticker.history(start="1950-01-01", end="2025-01-01", interval="1wk") # time period and weekly returns
close = data['Close']

# Calculate log returns
log_ret = np.log(close / close.shift(1)).dropna()

# Put Parameters
strike_ratio = 0.97 # Modified to 1 for full protection

premium_ratio = 0.0002 # Modify here if you want to include a premium

# Calculation for hedged portfolio log returns
strike = strike_ratio * close
premium_put = premium_ratio * close

payoff_put = np.maximum(strike - close.shift(-1), 0)
final_with_put = close.shift(-1) + payoff_put - premium_put
final_with_put = final_with_put.dropna()

log_ret_with_put = np.log(final_with_put / close.iloc[:-1].values)
log_ret_with_put = pd.Series(log_ret_with_put, index=final_with_put.index)

# Stress Test
# Calculation of log returns after stress test
extreme_moves = [-0.05, -0.3, -0.2, -0.1, -0.2, -0.3, -0.05, -0.3, -0.2, -0.1, -0.2, -0.3]
last_price = close.iloc[-1]
new_prices = [last_price * (1 + extreme_moves[0])]
for r in extreme_moves[1:]:
    new_prices.append(new_prices[-1] * (1 + r))
new_dates = [data.index[-1] + pd.Timedelta(weeks=i) for i in range(1, len(new_prices) + 1)]
data_ext = pd.DataFrame({'Close': new_prices}, index=new_dates)
data_stress = pd.concat([data, data_ext])

close_stress = data_stress['Close']
log_ret_stress = np.log(close_stress / close_stress.shift(1)).dropna()

strike_stress = strike_ratio * close_stress
payoff_put_stress = np.maximum(strike_stress - close_stress.shift(-1), 0)
premium_put_stress = premium_ratio * close_stress

final_with_put_stress = close_stress.shift(-1) + payoff_put_stress - premium_put_stress
final_with_put_stress = final_with_put_stress.dropna()
log_ret_with_put_stress = np.log(final_with_put_stress / close_stress.iloc[:-1].values)
log_ret_with_put_stress = pd.Series(log_ret_with_put_stress, index=final_with_put_stress.index)

print(log_ret_with_put_stress)

# Synthetic function for metrics
def stats(series):
    return {
        'Mean': series.mean(),

```

```

        'Std': series.std(),
        'Variance': series.var(),
        'Quantile_0.05': series.quantile(0.05)
    }

# Create summary DataFrame
df_summ = pd.DataFrame([
    stats(log_ret),
    stats(log_ret_with_put),
    stats(log_ret_stress),
    stats(log_ret_with_put_stress)
],
    index=['Orig_NonProt', 'Orig_Put', 'Stress_NonProt', 'Stress_Put']
)

print(df_summ)

#####
#                                     V ANALYSIS RESULTS
#####

# Calculation of functions
Omega = 0
Delta_s = 1e-3

# Calculation of specific K quantiles
K_orig = -0.032 # modify this K to replicate results

# Calculation of s^- for each distribution
s_orig = s_minus(log_ret, Omega)
s_put_orig = s_minus(log_ret_with_put, Omega)
s_nohedge = s_minus(log_ret_stress, Omega)
s_put_stress = s_minus(log_ret_with_put_stress, Omega)

# Calculation of lambda without common baseline
lambda_orig = lambda_from_s_empirical(s_orig, s_orig)
lambda_put_orig = lambda_from_s_empirical(s_put_orig, s_orig)
lambda_nohedge = lambda_from_s_empirical(s_nohedge, s_orig)
lambda_put_stress = lambda_from_s_empirical(s_put_stress, s_orig)

# Calculation of xi_K and Vega with specific K and personalized lambda
xi_orig = xi_K_on_baseline(log_ret, lambda_orig, Omega, K_orig)
V_orig = vega_tail_baseline(log_ret, s_orig, s_orig, Omega, K_orig, Delta_s)

xi_put_orig = xi_K_on_baseline(log_ret_with_put, lambda_put_orig, Omega, K_orig)
V_put_orig = vega_tail_baseline(log_ret_with_put, s_put_orig, s_orig, Omega, K_orig, Delta_s)

xi_nohedge = xi_K_on_baseline(log_ret_stress, lambda_nohedge, Omega, K_orig)
V_nohedge = vega_tail_baseline(log_ret_stress, s_nohedge, s_orig, Omega, K_orig, Delta_s)

xi_put_stress = xi_K_on_baseline(log_ret_with_put_stress, lambda_put_stress, Omega, K_orig)
V_put_stress = vega_tail_baseline(log_ret_with_put_stress, s_put_stress, s_orig, Omega, K_orig, Delta_s)

# Creation of summary DataFrame
df_metrics_nobase = pd.DataFrame({
    'Portfolio': [
        'Original unprotected',
        'Original with Put',
        'Stress test unprotected',
        'Stress test with Put'
    ],
    'Semi-deviation s^-': [

```

```

s_orig,
s_put_orig,
s_nohedge,
s_put_stress
],
'Lambda': [
lambda_orig,
lambda_put_orig,
lambda_nohedge,
lambda_put_stress
],
'Xi(K, s^-)': [
xi_orig,
xi_put_orig,
xi_nohedge,
xi_put_stress
],
'Vega': [
V_orig,
V_put_orig,
V_nohedge,
V_put_stress
]
})

print("0.05 Quantile for the portfolio without put", K_orig)
print("")
with pd.option_context('display.float_format', '{:,.6f}'.format):
    print(df_metrics_nobase.to_string(index=False))

#####
#                               W ANALYSIS RESULTS                               #
#####

Omega = 0
Delta_s = 1e-3

# Original baseline returns
baseline_returns = log_ret.copy()

# Calculation of single baseline L and H (those of the original portfolio)
L_baseline = 0.03 # modify this to replicate results
H_baseline = log_ret.max()

# Calculation of s_plus for original and put portfolio
s_plus_orig = s_plus(log_ret, Omega)
s_plus_put = s_plus(log_ret_with_put, Omega)

# Calculation of lambda+ relative to original s_plus baseline
lambda_plus_orig = lambda_from_s_plus_empirical(s_plus_orig, s_plus_orig)
lambda_plus_put = lambda_from_s_plus_empirical(s_plus_put, s_plus_orig)

# Calculation of xi+ for the two portfolios with fixed L_baseline and H_baseline
xi_plus_orig = xi_plus_on_baseline(baseline_returns, lambda_plus_orig, Omega, L_baseline, H_baseline)
xi_plus_put = xi_plus_on_baseline(baseline_returns, lambda_plus_put, Omega, L_baseline, H_baseline)

# Calculation of sensitivity W with fixed baseline L and H
W_orig = w_vega_right_baseline(log_ret, s_plus_orig, s_plus_orig, Omega, L_baseline, H_baseline, Delta_s)
W_put = w_vega_right_baseline(log_ret_with_put, s_plus_put, s_plus_orig, Omega, L_baseline, H_baseline, Delta_s)

# Creation of summary DataFrame without stress test
df_metrics_right_simple = pd.DataFrame({

```

```

'Portfolio': [
    'Original unprotected',
    'Original with Put'
],
'Semi-deviation s^+': [
    s_plus_orig,
    s_plus_put
],
'Lambda^+': [
    lambda_plus_orig,
    lambda_plus_put
],
'Xi^+(L,H)': [
    xi_plus_orig,
    xi_plus_put
],
'W (Vega Sensitivity s^+)': [
    W_orig,
    W_put
]
})
with pd.option_context('display.float_format', '{:,.6f}'.format):
    print(df_metrics_right_simple.to_string(index=False))

#####
#                BARBELL PORTFOLIOS SINGLE ASSET ANALYSIS
#####

# Instruments to include in the portfolio
assets = ['AAPL', 'MSFT', 'GOOGL', 'AMZN']
crypto_candidates = ['BTC-USD', 'ETH-USD', 'LTC-USD', 'XRP-USD', 'DOGE-USD']
smallcap_candidates = ['NIO', 'TRMD', 'FUBO', 'AAOI', 'SPCE', 'ZM']
start = "2020-01-01"
end = "2025-01-01"
min_weeks = 5 * 52 # ~260 weeks

# download weekly history and calculate log returns
def weekly_log_returns(ticker, start, end):
    t = yf.Ticker(ticker)
    hist = t.history(start=start, end=end, interval="1 wk")[['Close']].rename(columns={'Close': ticker})
    if hist.empty:
        return pd.Series(dtype=float)
    # Normalize index (remove timezones)
    hist.index = pd.to_datetime(hist.index)
    try:
        hist.index = hist.index.tz_convert(None)
    except (TypeError, AttributeError):
        try:
            hist.index = hist.index.tz_localize(None)
        except Exception:
            pass
    lr = np.log(hist[ticker] / hist[ticker].shift(1)).dropna()
    lr.name = ticker
    return lr

# Download weekly returns
all_candidates = assets + crypto_candidates + smallcap_candidates
returns = {}

for tk in all_candidates:
    try:
        lr = weekly_log_returns(tk, start, end)

```

```

        if len(lr) >= min_weeks:
            returns[tk] = lr
        else:
            print(f"Discarded {tk}: {len(lr)} weeks (< {min_weeks})")
    except Exception as e:
        print(f"Error downloading {tk}: {e}")

if not returns:
    raise RuntimeError("No tickers with at least 5 years of weekly data.")

# Build DataFrame
df_rets = pd.concat(returns.values(), axis=1)
df_rets.columns = list(returns.keys())
df_rets.sort_index(inplace=True)

# Normalize index and resolve timezones / duplicates
df_rets.index = pd.to_datetime(df_rets.index)
try:
    df_rets.index = df_rets.index.tz_convert(None)
except (TypeError, AttributeError):
    try:
        df_rets.index = df_rets.index.tz_localize(None)
    except Exception:
        pass

# Resample on Thursday to uniformize all series
df_rets = df_rets.resample('W-THU').last()

# --- 3) Risk-free rate (from ^IRX) ---
print("Downloading risk-free rate (^IRX)...")
irx = yf.Ticker("^IRX").history(start=start, end=end, interval="1wk")['Close']
irx.index = pd.to_datetime(irx.index).tz_localize(None)
rf_for_weeks = ((1 + irx / 100) ** (1 / 52) - 1).rename('rf_weekly')
rf_for_weeks = rf_for_weeks.reindex(df_rets.index, method='ffill')
print("Weekly risk-free rate example:")
print(rf_for_weeks.head())

# Preparation for real portfolio
asset_simple = np.exp(df_rets) - 1

# Align the risk-free as an additional asset
rf_aligned = rf_for_weeks.reindex(df_rets.index).ffill()
df_rets_with_rf = pd.concat([df_rets, rf_aligned.rename('RF')], axis=1)

# --- 5) Final Save ---
final_df = df_rets_with_rf.dropna(how='all')
print("Final DataFrame head (log returns + risk-free):")
print(final_df.head())

final_df.to_csv("weekly_log_returns_with_rf.csv")

#####
#                                     CALCULATION OF V AND W FOR EACH ASSET
#####

asset_tail_metrics_df = pd.DataFrame(index=final_df.columns,
                                     columns=['s_minus', 'xi', 'Vega', 's_plus', 'xi_plus', 'W'])

Omega = 0
quantile = 0.05
Delta_s = 1e-2

```

```

for asset in final_df.columns:
    data = final_df[asset].dropna()
    if data.empty:
        continue

    # Left tail
    K = -0.03 # change this to replicate results
    s_minus_val = s_minus(data, Omega)
    lambda_minus = lambda_from_s_empirical(s_minus_val, s_minus_val)
    xi_minus_val = xi_K_on_baseline(data, lambda_minus, Omega, K)
    V_val = vega_tail_baseline(data, s_minus_val, s_minus_val, Omega, K, Delta_s)

    # Right tail
    L = 0.03 # change this to replicate results
    s_plus_val = s_plus(data, Omega)
    lambda_plus = lambda_from_s_plus_empirical(s_plus_val, s_plus_val)
    xi_plus_val = xi_plus_on_baseline(data, lambda_plus, Omega, L, data.max())
    W_val = w_vega_right_baseline(data, s_plus_val, s_plus_val, Omega, L, data.max(), Delta_s=1e-2)

    asset_tail_metrics_df.loc[asset] = [s_minus_val, xi_minus_val, V_val, s_plus_val, xi_plus_val, W_val]

asset_tail_metrics_df = asset_tail_metrics_df

print("")
print("Tail metrics for each asset:")
print("")
print(asset_tail_metrics_df.round(4))

#####
#                                     BARBELL PORTFOLIO 85% RF
#####

rf_weight = 0.85          # 85% capital in risk-free
risk_asset_weight = 1 - rf_weight
min_weight_per_asset = 0.01 # CONSTRAINT: at least 1% for each risky asset

# Identify risky asset columns (excluding 'RF')
risky_assets_cols = [col for col in final_df.columns if col != 'RF']

# Redefine data with risky assets only
assets_returns = final_df[risky_assets_cols]
mu = assets_returns.mean()
Sigma = assets_returns.cov()
n_assets = len(assets_returns.columns) # CORRECT: n_assets = 15

# Keep RF series separate
rf_series = final_df['RF']

# PORTFOLIO CONSTRUCTION
weights_minvar = solve_portfolio(constraints_minvar, objective_minvar)
weights_minvar['RF'] = rf_weight

# HighRisk (ex Barbell)
vols = assets_returns.std().sort_values(ascending=False)
top_risky = vols.index[:5]
weights_highrisk = pd.Series(0.0, index=assets_returns.columns)
weights_highrisk[top_risky] = risk_asset_weight / len(top_risky)
weights_highrisk['RF'] = rf_weight

# Equal Weight
weights_equal = pd.Series(risk_asset_weight / n_assets, index=assets_returns.columns)
weights_equal['RF'] = rf_weight

```

```

# Target risk/return portfolios (SR)
target_returns = [0.005, 0.007, 0.010] # low, medium, high
weights_sr_list = []

for i, tr in enumerate(target_returns, 1):
    def constraints_sr(w):
        return [cp.sum(w) == 1,
                w >= min_weight_per_asset,
                cp.sum(cp.multiply(mu.values, w)) >= tr]
    w_sr = solve_portfolio(constraints_sr, objective_minvar)
    w_sr['RF'] = rf_weight
    weights_sr_list.append((f"SR_{i}", w_sr))

# Final portfolios DataFrame
portfolios_dict = {
    'MinVar': weights_minvar,
    'HighRisk': weights_highrisk,
    'EqualWeight': weights_equal
}
for name, w_sr in weights_sr_list:
    portfolios_dict[name] = w_sr

barbell_85_rf = pd.DataFrame(portfolios_dict)
barbell_85_rf = barbell_85_rf[sorted(barbell_85_rf.columns)]
print("Barbell portfolios 85% RF:")
print(barbell_85_rf.round(4))

# Assuming 'barbell_85_rf' is your weights DataFrame
sum_weights = barbell_85_rf.sum()
print("Sum of weights for each portfolio:")
print(sum_weights.round(8))

# CALCULATION OF HISTORICAL RETURNS SERIES
port_returns_df = pd.DataFrame(index=final_df.index)

for col in barbell_85_rf.columns:
    weights = barbell_85_rf[col]
    port_returns_df[col] = (assets_returns * weights[assets_returns.columns]).sum(axis=1) + weights['RF'] * rf_series

print("\nHistorical returns series of portfolios:")
print(port_returns_df.head())

# Save portfolio returns in a CSV file
port_returns_df.to_csv('portfolio_returns_85RF.csv')

# TAIL METRICS
tail_metrics_df = pd.DataFrame(index=barbell_85_rf.columns,
                                columns=['s_minus', 'xi', 'Vega', 's_plus', 'xi_plus', 'W'])

Omega = 0
K, L, Delta_s = -0.01, 0.01, 1e-3 # CHANGE THESE TO REPLICATE RESULTS

Delta_s = 1e-2
for port in barbell_85_rf.columns:
    data = port_returns_df[port]
    s_minus_val = s_minus(data, Omega)
    lambda_minus = lambda_from_s_empirical(s_minus_val, s_minus_val)
    xi_minus_val = xi_K_on_baseline(data, lambda_minus, Omega, K)
    V_val = vega_tail_baseline(data, s_minus_val, s_minus_val, Omega, K, Delta_s)
    s_plus_val = s_plus(data, Omega)
    lambda_plus = lambda_from_s_plus_empirical(s_plus_val, s_plus_val)

```

```

xi_plus_val = xi_plus_on_baseline(data, lambda_plus, Omega, L, data.max())
W_val = w_vega_right_baseline(data, s_plus_val, s_plus_val, Omega, L, data.max(), Delta_s)
tail_metrics_df.loc[port] = [s_minus_val, xi_minus_val, V_val, s_plus_val, xi_plus_val, W_val]

print("\n Tail metrics for portfolios (85% RF):")
print(tail_metrics_df.round(4))

#####
#                                     BARBELL PORTFOLIO 90% RF
#####

rf_weight = 0.90          # 90% capital in risk-free
risk_asset_weight = 1 - rf_weight
min_weight_per_asset = 0.01 # at least 1% for each risky asset

# Identify risky asset columns (excluding 'RF')
risky_assets_cols = [col for col in final_df.columns if col != 'RF']

# Redefine data with risky assets only
assets_returns = final_df[risky_assets_cols]
mu = assets_returns.mean()
Sigma = assets_returns.cov()
n_assets = len(assets_returns.columns) # CORRECT: n_assets = 15

# Keep RF series separate
rf_series = final_df['RF']

# PORTFOLIO CONSTRUCTION
weights_minvar = solve_portfolio(constraints_minvar, objective_minvar)
weights_minvar['RF'] = rf_weight

# HighRisk (ex Barbell)
vols = assets_returns.std().sort_values(ascending=False)
top_risky = vols.index[:5]
weights_highrisk = pd.Series(0.0, index=assets_returns.columns)
weights_highrisk[top_risky] = risk_asset_weight / len(top_risky)
weights_highrisk['RF'] = rf_weight

# Equal Weight
weights_equal = pd.Series(risk_asset_weight / n_assets, index=assets_returns.columns)
weights_equal['RF'] = rf_weight

# Target risk/return portfolios (SR)
target_returns = [0.005, 0.007, 0.010] # low, medium, high
weights_sr_list = []

for i, tr in enumerate(target_returns, 1):
    def constraints_sr(w):
        return [cp.sum(w) == 1,
                w >= min_weight_per_asset,
                cp.sum(cp.multiply(mu.values, w)) >= tr]
    w_sr = solve_portfolio(constraints_sr, objective_minvar)
    w_sr['RF'] = rf_weight
    weights_sr_list.append((f'SR_{i}', w_sr))

# Final portfolios DataFrame
portfolios_dict = {
    'MinVar': weights_minvar,
    'HighRisk': weights_highrisk,
    'EqualWeight': weights_equal
}
for name, w_sr in weights_sr_list:

```

```

portfolios_dict[name] = w_sr

barbell_90_rf = pd.DataFrame(portfolios_dict)
barbell_90_rf = barbell_90_rf[sorted(barbell_90_rf.columns)]
print("Barbell portfolios 90% RF:")
print(barbell_90_rf.round(4))

# Assuming 'barbell_85_rf' is your weights DataFrame
sum_weights = barbell_90_rf.sum()
print("")
print("Sum of weights for each portfolio:")
print(sum_weights.round(8))

# CALCULATION OF HISTORICAL RETURNS SERIES

port_returns_df = pd.DataFrame(index=final_df.index)

for col in barbell_90_rf.columns:
    weights = barbell_90_rf[col]
    port_returns_df[col] = (assets_returns * weights[assets_returns.columns]).sum(axis=1) + weights['RF'] * rf_series

print("\nHistorical returns series of portfolios:")
print(port_returns_df.head())

# Save portfolio returns in a CSV file
port_returns_df.to_csv('portfolio_returns_90RF.csv')

# TAIL METRICS
tail_metrics_df = pd.DataFrame(index=barbell_90_rf.columns,
                                columns=['s_minus', 'xi', 'Vega', 's_plus', 'xi_plus', 'W'])

Omega = 0
K, L, Delta_s = -0.01, 0.01, 1e-3 # CHANGE THESE TO REPLICATE RESULTS

for port in barbell_90_rf.columns:
    data = port_returns_df[port]
    s_minus_val = s_minus(data, Omega)
    lambda_minus = lambda_from_s_empirical(s_minus_val, s_minus_val)
    xi_minus_val = xi_K_on_baseline(data, lambda_minus, Omega, K)
    V_val = vega_tail_baseline(data, s_minus_val, s_minus_val, Omega, K, Delta_s)
    s_plus_val = s_plus(data, Omega)
    lambda_plus = lambda_from_s_plus_empirical(s_plus_val, s_plus_val)
    xi_plus_val = xi_plus_on_baseline(data, lambda_plus, Omega, L, data.max())
    W_val = w_vega_right_baseline(data, s_plus_val, s_plus_val, Omega, L, data.max(), Delta_s)
    tail_metrics_df.loc[port] = [s_minus_val, xi_minus_val, V_val, s_plus_val, xi_plus_val, W_val]

print("\n Tail metrics for portfolios (90% RF):")
print(tail_metrics_df.round(4))

#####
#                                     BARBELL PORTFOLIO 95% RF
#####

rf_weight = 0.95          # 95% capital in risk-free
risk_asset_weight = 1 - rf_weight
min_weight_per_asset = 0.01 # at least 1% for each risky asset

# Identify risky asset columns (excluding 'RF')
risky_assets_cols = [col for col in final_df.columns if col != 'RF']

# Redefine data with risky assets only
assets_returns = final_df[risky_assets_cols]

```

```

mu = assets_returns.mean()
Sigma = assets_returns.cov()
n_assets = len(assets_returns.columns) # CORRECT: n_assets = 15

# Keep RF series separate
rf_series = final_df['RF']

# PORTFOLIO CONSTRUCTION
weights_minvar = solve_portfolio(constraints_minvar, objective_minvar)
weights_minvar['RF'] = rf_weight

# HighRisk (ex Barbell)
vols = assets_returns.std().sort_values(ascending=False)
top_risky = vols.index[:5]
weights_highrisk = pd.Series(0.0, index=assets_returns.columns)
weights_highrisk[top_risky] = risk_asset_weight / len(top_risky)
weights_highrisk['RF'] = rf_weight

# Equal Weight
weights_equal = pd.Series(risk_asset_weight / n_assets, index=assets_returns.columns)
weights_equal['RF'] = rf_weight

# Target risk/return portfolios (SR)
target_returns = [0.005, 0.007, 0.010]
weights_sr_list = []

for i, tr in enumerate(target_returns, 1):
    def constraints_sr(w):
        return [cp.sum(w) == 1, w >= min_weight_per_asset, cp.sum(cp.multiply(mu.values, w)) >= tr]
    w_sr = solve_portfolio(constraints_sr, objective_minvar)
    w_sr['RF'] = rf_weight
    weights_sr_list.append((f'SR_{i}', w_sr))

portfolios_dict = {
    'MinVar': weights_minvar,
    'HighRisk': weights_highrisk,
    'EqualWeight': weights_equal
}
for name, w_sr in weights_sr_list:
    portfolios_dict[name] = w_sr

barbell_95_rf = pd.DataFrame(portfolios_dict)
barbell_95_rf = barbell_95_rf[sorted(barbell_95_rf.columns)]
print("Barbell portfolios 95% RF:")
print(barbell_95_rf.round(4))

# Assuming 'barbell_85_rf' is your weights DataFrame
sum_weights = barbell_95_rf.sum()
print("")
print("Sum of weights for each portfolio:")
print(sum_weights.round(8))

# CALCULATION OF HISTORICAL RETURNS SERIES
port_returns_df = pd.DataFrame(index=final_df.index)

for col in barbell_95_rf.columns:
    weights = barbell_95_rf[col]
    port_returns_df[col] = (assets_returns * weights[assets_returns.columns]).sum(axis=1) + weights['RF'] * rf_series

print("\nHistorical returns series of portfolios:")
print(port_returns_df.head())
# Save portfolio returns in a CSV file

```

```

port_returns_df.to_csv('portfolio_returns_95RF.csv')

# TAIL METRICS
tail_metrics_df = pd.DataFrame(index=barbell_95_rf.columns,
                               columns=['s_minus', 'xi', 'Vega', 's_plus', 'xi_plus', 'W'])

Omega = 0
K, L, Delta_s = -0.01, 0.01, 1e-3 # CHANGE THESE TO REPLICATE RESULTS

for port in barbell_95_rf.columns:
    data = port_returns_df[port]
    s_minus_val = s_minus(data, Omega)
    lambda_minus = lambda_from_s_empirical(s_minus_val, s_minus_val)
    xi_minus_val = xi_K_on_baseline(data, lambda_minus, Omega, K)
    V_val = vega_tail_baseline(data, s_minus_val, s_minus_val, Omega, K, Delta_s)
    s_plus_val = s_plus(data, Omega)
    lambda_plus = lambda_from_s_plus_empirical(s_plus_val, s_plus_val)
    xi_plus_val = xi_plus_on_baseline(data, lambda_plus, Omega, L, data.max())
    W_val = w_vega_right_baseline(data, s_plus_val, s_plus_val, Omega, L, data.max(), Delta_s)
    tail_metrics_df.loc[port] = [s_minus_val, xi_minus_val, V_val, s_plus_val, xi_plus_val, W_val]

print("\n Tail metrics for portfolios (95% RF):")
print(tail_metrics_df.round(4))

```

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